

AZ-104 Microsoft Azure Administrator Complete Master Practical Scenario Guide — 2026

211 practical scenarios with simple explanations, step-by-step solutions, exam points, troubleshooting, CLI/PowerShell practice, ARM/Bicep practice, and service-selection questions.

Scope note: This guide is aligned to Microsoft's AZ-104 Skills Measured objectives effective April 17, 2026. It is a preparation workbook, not a collection of real exam questions.

2026 Exam-Aligned Practical Preparation Workbook

Purpose: A hands-on scenario workbook based on Microsoft's current AZ-104 skills measured as of April 17, 2026. It expands the user's original 100-scenario guide into 100 unique exercises covering all five current exam domains, with step-by-step implementation and an exam-focused point for every scenario.

How to use this workbook

- Do each scenario yourself in an Azure subscription or sandbox before reading the solution.
- For every scenario, understand why the selected service/configuration is appropriate—not just the portal clicks.
- Practice the portal first, then repeat important tasks using Azure CLI, PowerShell, or ARM/Bicep.
- Delete lab resources after practice to control costs.
- After completing the workbook, use Microsoft's official practice assessment and exam sandbox.

Current AZ-104 exam coverage

Microsoft's current study guide lists five domains: Manage Azure identities and governance (20–25%), Implement and manage storage (15–20%), Deploy and manage Azure compute resources (20–25%), Implement and manage virtual networking (15–20%), and Monitor and maintain Azure resources (10–15%).

The current objectives explicitly include Entra users/groups and licenses, RBAC, Policy, locks, tags, subscriptions/costs; storage firewalls/SAS/access keys/object replication/data protection; ARM/Bicep, VMs/VMSS, ACI/Container Apps, App Service; VNets/peering/UDRs/NSGs/ASGs/Bastion/service endpoints/private endpoints/DNS/load balancing; and Azure Monitor, Network Watcher, Backup vaults, Site Recovery and failover.

Domain 1 — Manage Azure identities and governance (20–25%)

Scenario 1: Create a Microsoft Entra user and assign a license

Question / Business requirement: Create a Microsoft Entra user and assign a license. Perform the configuration and verify the expected outcome.

In simple words: We are creating a user account and giving that user the Microsoft service access they need.

Step-by-step solution

- Azure portal → Microsoft Entra ID → Users → New user.
- Enter identity details and create the user.
- Open the user → Licenses → Assignments → select the required license → Save.
- Verify the license appears under the user's Licenses blade.

AZ-104 exam point: Know the difference between an Entra identity and an Azure RBAC assignment.

Scenario 2: Create a security group and add members

Question / Business requirement: Create a security group and add members. Perform the configuration and verify the expected outcome.

In simple words: We are grouping users together so that permissions and access can be managed for the group instead of one user at a time.

Step-by-step solution

- Microsoft Entra ID → Groups → New group.
- Set Group type to Security, provide a name, and choose membership type.
- Create the group.
- Open the group → Members → Add members → select users → Add.

AZ-104 exam point: Know security groups versus Microsoft 365 groups and assigned versus dynamic membership.

Scenario 3: Configure a dynamic user group

Question / Business requirement: Configure a dynamic user group. Perform the configuration and verify the expected outcome.

Step-by-step solution

- Microsoft Entra ID → Groups → New group.
- Choose Security and Dynamic User membership.
- Add a rule such as user.department equals Finance.
- Validate the rule syntax and create the group.
- Verify membership after directory evaluation.

AZ-104 exam point: Understand that dynamic membership is rule-driven rather than manually maintained.

Scenario 4: Configure Self-Service Password Reset for selected users

Question / Business requirement: Configure Self-Service Password Reset for selected users. Perform the configuration and verify the expected outcome.

Step-by-step solution

- Microsoft Entra ID → Password reset.
- Set Self service password reset to Selected.
- Choose the target group.
- Configure authentication methods and registration requirements.
- Save and test with a member account.

AZ-104 exam point: Know the scope options: None, Selected, and All.

Scenario 5: Invite an external guest user

Question / Business requirement: Invite an external guest user. Perform the configuration and verify the expected outcome.

Step-by-step solution

- Microsoft Entra ID → Users → New user → Invite external user.
- Enter the guest email address and invitation details.
- Send the invitation.
- Open the guest user and verify User type is Guest.

AZ-104 exam point: Know the distinction between member and guest users.

Scenario 6: Assign Reader at resource-group scope

Question / Business requirement: Assign Reader at resource-group scope. Perform the configuration and verify the expected outcome.

Step-by-step solution

- Open the target resource group → Access control (IAM).
- Select Add → Add role assignment.
- Choose Reader.
- Select the target user or group.
- Review scope and complete the assignment.

AZ-104 exam point: RBAC scope can be subscription, resource group, or resource; assignments inherit downward.

Scenario 7: Determine effective Azure RBAC access

Question / Business requirement: Determine effective Azure RBAC access. Perform the configuration and verify the expected outcome.

Step-by-step solution

- Open a subscription/resource group/resource → Access control (IAM).
- Review Check access.
- Select the user or group.
- Review direct and inherited role assignments.

- Identify which assignment grants the required action.

AZ-104 exam point: Do not confuse Azure RBAC with Microsoft Entra directory roles.

Scenario 8: Create a custom Azure RBAC role

Question / Business requirement: Create a custom Azure RBAC role. Perform the configuration and verify the expected outcome.

Step-by-step solution

- Open the subscription or resource group where the role will be defined.
- Open Access control (IAM) → Add → Add custom role.
- Create from scratch or clone a suitable role.
- Specify allowed actions, excluded actions, and assignable scopes.
- Create the role and then assign it to the required principal.

AZ-104 exam point: Custom roles control Azure resource operations; they are not Entra directory roles.

Scenario 9: Move resources into a resource group

Question / Business requirement: Move resources into a resource group. Perform the configuration and verify the expected outcome.

Step-by-step solution

- Identify the resources that support resource-group moves.
- Open the resource group → Move → Move to another resource group.
- Select the destination resource group.
- Validate dependencies and acknowledge warnings.
- Start the move and verify resource IDs and relationships afterward.

AZ-104 exam point: Know that move operations have resource-specific restrictions and can affect dependencies.

Scenario 10: Create a management-group hierarchy

Question / Business requirement: Create a management-group hierarchy. Perform the configuration and verify the expected outcome.

Step-by-step solution

- Search for Management groups.
- Create a top-level management group.
- Create child management groups for business units.
- Move subscriptions into the appropriate management groups.
- Verify policy and RBAC inheritance.

AZ-104 exam point: Management groups sit above subscriptions and are useful for governance at scale.

Scenario 11: Apply an Azure Policy to require tags

Question / Business requirement: Apply an Azure Policy to require tags. Perform the configuration and verify the expected outcome.

Step-by-step solution

- Azure Policy → Definitions → search for a built-in tag policy.
- Assign the policy at the required scope.
- Set parameters such as required tag name.
- Review the assignment and remediation options.
- Create/update a resource and verify the policy evaluates it.

AZ-104 exam point: Understand Policy effects such as Audit, Deny, Modify, and DeployIfNotExists.

Scenario 12: Create a policy initiative

Question / Business requirement: Create a policy initiative. Perform the configuration and verify the expected outcome.

Step-by-step solution

- Azure Policy → Definitions → Initiative definitions → Add.
- Add related policy definitions.
- Configure parameters and groupings.
- Create the initiative.
- Assign the initiative to the required management group, subscription, or resource group.

AZ-104 exam point: An initiative is a collection of policy definitions managed as one assignment.

Scenario 13: Use a resource lock

Question / Business requirement: Use a resource lock. Perform the configuration and verify the expected outcome.

Step-by-step solution

- Open the subscription, resource group, or resource.
- Settings → Locks → Add.
- Choose Delete or Read-only.
- Name the lock and save.
- Attempt the protected operation to validate the behavior.

AZ-104 exam point: Locks can block operations even when a user has a high RBAC role.

Scenario 14: Apply and verify resource tags

Question / Business requirement: Apply and verify resource tags. Perform the configuration and verify the expected outcome.

Step-by-step solution

- Open a resource or resource group → Tags.
- Add keys such as Environment and Owner.
- Save the tags.
- Use Azure Resource Graph/portal filtering to verify the tags.
- Apply the organization's tagging standard to new resources.

AZ-104 exam point: Tags are metadata and do not automatically inherit unless implemented through policy.

Scenario 15: Create a cost budget and alert

Question / Business requirement: Create a cost budget and alert. Perform the configuration and verify the expected outcome.

Step-by-step solution

- Cost Management + Billing → Budgets.
- Select the subscription/resource-group scope.
- Create a monthly budget.
- Set threshold percentages such as 80% and 100%.
- Configure alert recipients and save.

AZ-104 exam point: Budgets alert on forecast/actual spend but do not automatically stop resources.

Scenario 16: Use Azure Advisor to identify recommendations

Question / Business requirement: Use Azure Advisor to identify recommendations. Perform the configuration and verify the expected outcome.

Step-by-step solution

- Open Azure Advisor.

- Select the subscription.
- Review cost, security, reliability, performance, and operational-excellence recommendations.
- Open a recommendation and inspect affected resources.
- Record or implement the applicable recommendation.

AZ-104 exam point: Advisor recommendations are different from Azure Policy enforcement.

Scenario 17: Manage a subscription and resource-provider registration

Question / Business requirement: Manage a subscription and resource-provider registration. Perform the configuration and verify the expected outcome.

Step-by-step solution

- Open Subscriptions → select the subscription.
- Review resource providers under Resource providers.
- Search for the required provider.
- Select Register if it is not registered.
- Retry deployment after registration completes.

AZ-104 exam point: Provider registration is a common cause of deployment failures.

Scenario 18: Create an Azure budget with a resource-group scope

Question / Business requirement: Create an Azure budget with a resource-group scope. Perform the configuration and verify the expected outcome.

Step-by-step solution

- Cost Management + Billing → Budgets.
- Choose the resource group scope.
- Set the budget amount and period.
- Configure email/action notifications.
- Save and verify the budget scope.

AZ-104 exam point: Be able to distinguish subscription-wide and resource-group cost scope.

Scenario 19: Troubleshoot a denied deployment caused by Azure Policy

Question / Business requirement: Troubleshoot a denied deployment caused by Azure Policy. Perform the configuration and verify the expected outcome.

Step-by-step solution

- Open the failed deployment details.
- Identify the policy assignment causing the denial.
- Open Azure Policy → Compliance.
- Inspect the policy definition and assignment parameters.
- Choose a compliant resource configuration or request an authorized policy exemption.

AZ-104 exam point: Policy denial is governance enforcement; it is not the same as RBAC authorization failure.

Domain 2 — Implement and manage storage (15–20%)

Scenario 20: Create a storage account for general-purpose workloads

Question / Business requirement: Create a storage account for general-purpose workloads. Perform the configuration and verify the expected outcome.

Step-by-step solution

Storage accounts → Create.

- Choose the appropriate subscription/resource group.
- Choose Standard performance and the required redundancy.

- Review networking, data protection, and encryption settings.
- Create the account and verify its properties.

AZ-104 exam point: Know when GPv2 is appropriate and how performance/redundancy choices affect features and cost.

Scenario 21: Configure storage redundancy

Question / Business requirement: Configure storage redundancy. Perform the configuration and verify the expected outcome.

Step-by-step solution

- Open the storage account → Data management/Redundancy.
- Review the current replication option.
- Select the required redundancy such as LRS, ZRS, GRS, or RA-GRS when supported.
- Save and verify the change.

AZ-104 exam point: Understand local, zone, and geo-redundancy and the read-access variants.

Scenario 22: Configure blob soft delete and container soft delete

Question / Business requirement: Configure blob soft delete and container soft delete. Perform the configuration and verify the expected outcome.

Step-by-step solution

Storage account → Data protection.

- Enable blob soft delete and choose retention.
- Enable container soft delete and choose retention.
- Save the configuration.
- Create/delete test data and verify recovery options.

AZ-104 exam point: Soft delete provides recoverability; it is not the same as versioning.

Scenario 23: Enable blob versioning

Question / Business requirement: Enable blob versioning. Perform the configuration and verify the expected outcome.

Step-by-step solution

Storage account → Data protection.

- Enable Blob versioning.
- Upload a blob and modify it.
- Open the blob and inspect versions.
- Restore or promote an earlier version when required.

AZ-104 exam point: Versioning preserves previous blob versions and works with other data-protection features.

Scenario 24: Configure blob lifecycle management

Question / Business requirement: Configure blob lifecycle management. Perform the configuration and verify the expected outcome.

Step-by-step solution

Storage account → Lifecycle management → Add rule.

- Define the blob scope/prefix or tags.
- Set conditions such as days since modification.
- Choose actions such as move to Cool/Cold/Archive or delete.
- Save and review the rule.

AZ-104 exam point: Know that lifecycle rules automate data movement/deletion based on conditions.

Scenario 25: Create an Azure Blob container

Question / Business requirement: Create an Azure Blob container. Perform the configuration and verify the expected outcome.

Step-by-step solution

- Open Storage account → Containers.
- Create a container with an appropriate name.
- Choose the required anonymous access setting.
- Create the container.
- Upload a test blob and verify access.

AZ-104 exam point: Know that public access should be disabled unless explicitly required.

Scenario 26: Generate a least-privilege SAS

Question / Business requirement: Generate a least-privilege SAS. Perform the configuration and verify the expected outcome.

Step-by-step solution

Storage account → Shared access signature.

- Select only the required services and resource types.
- Select only needed permissions such as Read.
- Set a short expiry and appropriate protocol restrictions.
- Generate the SAS and test the intended operation.

AZ-104 exam point: Use least privilege and short validity; know service SAS versus account SAS at a conceptual level.

Scenario 27: Configure a stored access policy for a container

Question / Business requirement: Configure a stored access policy for a container. Perform the configuration and verify the expected outcome.

Step-by-step solution

- Open the container and its access policy settings.
- Create a stored access policy.
- Specify permissions and expiry.
- Save the policy.
- Use the policy identifier when generating an appropriate SAS.

AZ-104 exam point: Stored access policies allow centralized control of associated SAS permissions.

Scenario 28: Rotate storage account access keys

Question / Business requirement: Rotate storage account access keys. Perform the configuration and verify the expected outcome.

Step-by-step solution

Storage account → Access keys.

- Identify which key is currently in use by applications.
- Regenerate the unused key.
- Update applications with the new key.
- Regenerate the old key after all consumers have migrated.

AZ-104 exam point: Never regenerate a key blindly if applications still depend on it.

Scenario 29: Restrict storage access with a firewall

Question / Business requirement: Restrict storage access with a firewall. Perform the configuration and verify the expected outcome.

Step-by-step solution

Storage account → Networking.

- Set network access to selected networks/IP addresses.
- Add approved VNets/subnets or public IP ranges.
- Configure trusted service exceptions only when required.
- Save and test access from an approved and unapproved source.

AZ-104 exam point: Storage firewall rules and private endpoints solve different connectivity requirements.

Scenario 30: Create a storage private endpoint

Question / Business requirement: Create a storage private endpoint. Perform the configuration and verify the expected outcome.

Step-by-step solution

Storage account → Networking → Private endpoint.

- Choose the target storage subresource such as blob.
- Select the VNet and subnet.
- Configure Private DNS integration when required.
- Create the endpoint and verify the private IP/DNS resolution.

AZ-104 exam point: Private endpoints provide private connectivity to PaaS resources.

Scenario 31: Configure a storage service endpoint

Question / Business requirement: Configure a storage service endpoint. Perform the configuration and verify the expected outcome.

Step-by-step solution

- Open the VNet/subnet.
- Open Service endpoints.
- Add Microsoft.Storage.
- Save the subnet configuration.
- Configure the storage firewall to allow that subnet and test access.

AZ-104 exam point: Service endpoints keep traffic on the Azure backbone but the storage service still has a public endpoint.

Scenario 32: Use Azure Storage Explorer

Question / Business requirement: Use Azure Storage Explorer. Perform the configuration and verify the expected outcome.

Step-by-step solution

- Install/open Azure Storage Explorer.
- Sign in with the appropriate identity.
- Connect to the storage account.
- Browse blobs/files/queues/tables as permitted.
- Upload/download a test object and verify permissions.

AZ-104 exam point: Know that Storage Explorer is a management client, not a storage service.

Scenario 33: Use AzCopy to transfer blobs

Question / Business requirement: Use AzCopy to transfer blobs. Perform the configuration and verify the expected outcome.

Step-by-step solution

- Open Cloud Shell or a machine with AzCopy.
- Authenticate with the supported identity/SAS method.
- Run an AzCopy copy/sync operation for the source and destination.
- Review transfer output.
- Verify the destination objects.

AZ-104 exam point: Know the difference between copy and sync semantics.

Scenario 34: Configure Azure Files identity-based access

Question / Business requirement: Configure Azure Files identity-based access. Perform the configuration and verify the expected outcome.

Step-by-step solution

Storage account → File shares.

- Configure the appropriate identity-based authentication integration.
- Create the file share.
- Configure share-level permissions.
- Assign appropriate directory/file permissions where applicable and test access.

AZ-104 exam point: Know the distinction between share-level Azure RBAC and directory/file-level permissions.

Scenario 35: Create an Azure Files snapshot

Question / Business requirement: Create an Azure Files snapshot. Perform the configuration and verify the expected outcome.

Step-by-step solution

- Open the file share.
- Select Snapshots → Add.
- Create the snapshot.
- Open the snapshot and verify files.
- Use the snapshot to recover required files.

AZ-104 exam point: Snapshots provide point-in-time recovery for Azure Files.

Scenario 36: Configure Azure Files soft delete

Question / Business requirement: Configure Azure Files soft delete. Perform the configuration and verify the expected outcome.

Step-by-step solution

Storage account → Data protection or file-share data protection settings.

- Enable soft delete for file shares.
- Set retention.
- Save.
- Delete a test share and verify it can be recovered during retention.

AZ-104 exam point: Know the purpose and retention behavior of soft delete.

Scenario 37: Configure blob object replication

Question / Business requirement: Configure blob object replication. Perform the configuration and verify the expected outcome.

Step-by-step solution

- Create compatible source and destination storage accounts.
- Configure the required data-protection prerequisites.

- Create an object replication policy.
- Select source/destination containers and matching rules.
- Upload a qualifying blob and verify replication status.

AZ-104 exam point: Object replication is for asynchronous replication of selected blob data, not full account geo-replication.

Scenario 38: Configure storage account encryption

Question / Business requirement: Configure storage account encryption. Perform the configuration and verify the expected outcome.

Step-by-step solution

- Open the storage account → Encryption.
- Review Microsoft-managed key settings.
- If required by the scenario, configure customer-managed key integration and its identity/key permissions.
- Save and verify encryption configuration.

AZ-104 exam point: Understand Microsoft-managed versus customer-managed keys and the permissions needed.

Scenario 39: Select the correct storage tier

Question / Business requirement: Select the correct storage tier. Perform the configuration and verify the expected outcome.

Step-by-step solution

- Identify the workload's access frequency and retention needs.
- Open the storage account/container/blob settings.
- Choose Hot, Cool, Cold, or Archive where supported.
- Save and verify the access tier.
- Consider retrieval and early-deletion implications before choosing Archive.

AZ-104 exam point: Exam questions often test cost/performance trade-offs between access tiers.

Domain 3 — Deploy and manage Azure compute resources (20–25%)

Scenario 40: Deploy a Windows VM

Question / Business requirement: Deploy a Windows VM. Perform the configuration and verify the expected outcome.

Step-by-step solution

- Virtual machines → Create.
- Select subscription/resource group, image, size, administrator credentials.
- Configure networking and inbound access appropriately.
- Review disks and management settings.
- Create the VM and verify connection.

AZ-104 exam point: Know the relationships among VM, NIC, subnet, NSG, public IP, and managed disks.

Scenario 41: Resize a virtual machine

Question / Business requirement: Resize a virtual machine. Perform the configuration and verify the expected outcome.

Step-by-step solution

- Open the VM → Size.
- Review available sizes for the region and current configuration.
- Select the required size.
- Apply the change, accepting restart requirements.

- Verify the new VM size.

AZ-104 exam point: Some size changes require deallocation and availability depends on region/quota.

Scenario 42: Add and initialize a managed data disk

Question / Business requirement: Add and initialize a managed data disk. Perform the configuration and verify the expected outcome.

Step-by-step solution

- VM → Disks → Create and attach a new disk.
- Choose the disk type and size.
- Save the VM configuration.
- Inside the guest OS, initialize/format the disk and assign a mount point/drive letter.
- Verify read/write access.

AZ-104 exam point: Know OS disk versus data disk and the effect of disk performance tiers.

Scenario 43: Move a VM to another resource group

Question / Business requirement: Move a VM to another resource group. Perform the configuration and verify the expected outcome.

Step-by-step solution

- Open the VM → Move → Move to another resource group.
- Select the destination resource group.
- Review dependent resources.
- Validate and start the move.
- Verify the VM and supported dependencies in the destination.

AZ-104 exam point: Resource moves have constraints; always review validation results.

Scenario 44: Move a VM to another subscription

Question / Business requirement: Move a VM to another subscription. Perform the configuration and verify the expected outcome.

Step-by-step solution

- Open the VM/resource group and select Move.
- Choose Move to another subscription.
- Select the target subscription/resource group.
- Validate resource dependencies and provider requirements.
- Confirm the move and verify the target subscription.

AZ-104 exam point: Some resource types and configurations cannot be moved across subscriptions.

Scenario 45: Deploy a VM across Availability Zones

Question / Business requirement: Deploy a VM across Availability Zones. Perform the configuration and verify the expected outcome.

Step-by-step solution

- Create a VM in a region supporting zones.
- Choose an Availability Zone during VM creation.
- Deploy additional instances in different zones.
- Configure the application/load balancer to distribute traffic.
- Test zonal resilience.

AZ-104 exam point: Zones protect against datacenter-level failures within a region.

Scenario 46: Create an Availability Set

Question / Business requirement: Create an Availability Set. Perform the configuration and verify the expected outcome.

Step-by-step solution

- Search Availability sets → Create.
- Choose the region and appropriate configuration.
- Create multiple VMs in the same availability set.
- Verify fault/update domain placement.
- Test maintenance/failure behavior conceptually.

AZ-104 exam point: Availability Sets and Availability Zones are different resilience mechanisms.

Scenario 47: Create a VM Scale Set with autoscale

Question / Business requirement: Create a VM Scale Set with autoscale. Perform the configuration and verify the expected outcome.

Step-by-step solution

- Virtual machine scale sets → Create.
- Configure image, instance count, networking and identity.
- Open Scaling and configure minimum/default/maximum instances.
- Create rules based on metrics such as CPU.
- Save and verify scale-out/scale-in behavior.

AZ-104 exam point: Understand autoscale boundaries and scale-in/scale-out triggers.

Scenario 48: Enable VM boot diagnostics

Question / Business requirement: Enable VM boot diagnostics. Perform the configuration and verify the expected outcome.

Step-by-step solution

- Open the VM → Boot diagnostics.
- Enable the feature and choose the supported storage/configuration option.
- Save.
- Restart or reproduce the startup issue.
- Review the boot screenshot and serial/diagnostic information.

AZ-104 exam point: Boot diagnostics helps troubleshoot startup problems.

Scenario 49: Configure VM extension

Question / Business requirement: Configure VM extension. Perform the configuration and verify the expected outcome.

Step-by-step solution

- Open the VM → Extensions + applications.
- Select an appropriate extension such as Custom Script or monitoring extension.
- Provide required settings/script.
- Install and wait for provisioning.
- Verify extension status and resulting configuration.

AZ-104 exam point: Extensions run guest-level configuration tasks and can fail independently of the VM resource.

Scenario 50: Enable encryption at host

Question / Business requirement: Enable encryption at host. Perform the configuration and verify the expected outcome.

Step-by-step solution

- Open the VM configuration/security settings.
- Verify the selected VM size/region supports the capability.
- Enable encryption at host.
- Save/apply the configuration.
- Verify the setting on the VM.

AZ-104 exam point: Know that encryption at host differs from disk encryption.

Scenario 51: Deploy an ARM template

Question / Business requirement: Deploy an ARM template. Perform the configuration and verify the expected outcome.

Step-by-step solution

- Open a JSON ARM template with parameters.
- Azure portal → Deploy a custom template.
- Load the template.
- Provide parameters and target resource group.
- Review + create and verify the deployment.

AZ-104 exam point: Know parameters, variables, resources, dependencies, and outputs at a conceptual level.

Scenario 52: Deploy a Bicep file

Question / Business requirement: Deploy a Bicep file. Perform the configuration and verify the expected outcome.

Step-by-step solution

- Open Cloud Shell or use a supported deployment workflow.
- Select the subscription/resource group.
- Run an ARM/Bicep deployment command using the Bicep file.
- Provide required parameters.
- Verify deployment resources and outputs.

AZ-104 exam point: Understand that Bicep is a declarative language that compiles to ARM template JSON.

Scenario 53: Convert an ARM template to Bicep

Question / Business requirement: Convert an ARM template to Bicep. Perform the configuration and verify the expected outcome.

Step-by-step solution

- Open Cloud Shell.
- Use the Bicep CLI decompile capability on the ARM template.
- Review generated Bicep.
- Correct or simplify any generated expressions.
- Deploy the Bicep file to validate it.

AZ-104 exam point: Generated Bicep may need review before production use.

Scenario 54: Create an Azure Container Registry

Question / Business requirement: Create an Azure Container Registry. Perform the configuration and verify the expected outcome.

Step-by-step solution

- Container registries → Create.
- Select SKU and networking settings.
- Create the registry.

- Authenticate with an approved identity or method.
- Push and list a test image.

AZ-104 exam point: Know Basic/Standard/Premium capabilities conceptually and the importance of secure authentication.

Scenario 55: Deploy an Azure Container Instance

Question / Business requirement: Deploy an Azure Container Instance. Perform the configuration and verify the expected outcome.

Step-by-step solution

- Container instances → Create.
- Specify image, container ports, CPU/memory and restart policy.
- Configure networking/environment variables as required.
- Create the container group.
- Review logs and container state.

AZ-104 exam point: ACI is suited to simple container execution without managing VM clusters.

Scenario 56: Deploy an Azure Container App

Question / Business requirement: Deploy an Azure Container App. Perform the configuration and verify the expected outcome.

Step-by-step solution

- Container Apps → Create.
- Select/create an environment.
- Specify the container image and resources.
- Configure ingress and scaling rules.
- Create the app and verify revisions and scaling behavior.

AZ-104 exam point: Know that Container Apps provides managed application/container features beyond basic ACI.

Scenario 57: Configure an App Service plan and scale

Question / Business requirement: Configure an App Service plan and scale. Perform the configuration and verify the expected outcome.

Step-by-step solution

- App Service plans → Create.
- Choose OS, region and pricing tier.
- Create an App Service under the plan.
- Open Scale up or Scale out.
- Configure the required manual/autoscale settings and verify.

AZ-104 exam point: App Service plan determines compute resources and pricing tier for its apps.

Scenario 58: Configure an App Service custom domain and TLS

Question / Business requirement: Configure an App Service custom domain and TLS. Perform the configuration and verify the expected outcome.

Step-by-step solution

- App Service → Custom domains → Add custom domain.
- Create/modify DNS records required for validation.
- Validate ownership.
- Add/import the appropriate TLS certificate.
- Bind the certificate to the hostname and verify HTTPS.

AZ-104 exam point: Know DNS validation and certificate binding concepts.

Scenario 59: Configure App Service deployment slots and swap

Question / Business requirement: Configure App Service deployment slots and swap. Perform the configuration and verify the expected outcome.

Step-by-step solution

- App Service → Deployment slots → Add slot.
- Configure the staging slot.
- Deploy/test the application in staging.
- Use Swap to exchange staging and production.
- Verify slot-specific settings and application behavior.

AZ-104 exam point: Know which settings can be slot-sticky and why slots reduce deployment risk.

Scenario 60: Configure App Service backup

Question / Business requirement: Configure App Service backup. Perform the configuration and verify the expected outcome.

Step-by-step solution

- App Service → Backups.
- Configure a storage destination.
- Set backup schedule and retention.
- Choose application/configuration data as supported.
- Run a backup and verify its status.

AZ-104 exam point: Backups require appropriate storage configuration and supported App Service tier/features.

Domain 4 — Implement and manage virtual networking (15-20%)

Scenario 61: Create a VNet with multiple subnets

Question / Business requirement: Create a VNet with multiple subnets. Perform the configuration and verify the expected outcome.

Step-by-step solution

- Virtual networks → Create.
- Define address space.
- Create separate application/data/management subnets.
- Review subnet address ranges for overlap.
- Create and verify the VNet.

AZ-104 exam point: Understand address-space planning and subnet boundaries.

Scenario 62: Configure VNet peering

Question / Business requirement: Configure VNet peering. Perform the configuration and verify the expected outcome.

Step-by-step solution

- Open VNet A → Peerings → Add.
- Select VNet B.
- Enable virtual network access as required.
- Configure gateway transit/use remote gateway only when appropriate.
- Create/verify peering in both directions where required.

AZ-104 exam point: Peering is private connectivity between VNets; it is not transitive by default.

Scenario 63: Assign a static public IP

Question / Business requirement: Assign a static public IP. Perform the configuration and verify the expected outcome.

Step-by-step solution

- Public IP addresses → Create.
- Choose the required SKU and allocation settings.
- Create the IP.
- Associate it with a supported resource such as a NIC/load balancer.
- Verify the resource retains the expected public IP.

AZ-104 exam point: Know when public IP is required and when private connectivity is preferable.

Scenario 64: Create a UDR for internet egress through an NVA

Question / Business requirement: Create a UDR for internet egress through an NVA. Perform the configuration and verify the expected outcome.

Step-by-step solution

- Route tables → Create.
- Add route 0.0.0.0/0 with next hop Virtual appliance.
- Specify the NVA private IP.
- Associate the route table with the target subnet.
- Verify effective routes from a VM.

AZ-104 exam point: UDRs influence routing; NSGs control traffic separately.

Scenario 65: Troubleshoot effective routes

Question / Business requirement: Troubleshoot effective routes. Perform the configuration and verify the expected outcome.

Step-by-step solution

- Open the VM NIC → Effective routes.
- Identify the route selected for the destination.
- Compare system routes, BGP routes, and UDRs.
- Check whether the next hop is reachable.
- Correct the route table/peering/VPN configuration and retest.

AZ-104 exam point: Most-specific route selection is a key troubleshooting concept.

Scenario 66: Create an NSG and associate it to a subnet

Question / Business requirement: Create an NSG and associate it to a subnet. Perform the configuration and verify the expected outcome.

Step-by-step solution

- Network security groups → Create.
- Add inbound/outbound rules with priority, source, destination, port and action.
- Associate the NSG with the target subnet.
- Verify effective security rules on a NIC.

AZ-104 exam point: Know that NSGs can be associated with subnets and NICs.

Scenario 67: Use Application Security Groups

Question / Business requirement: Use Application Security Groups. Perform the configuration and verify the expected outcome.

Step-by-step solution

- Create an Application Security Group.

- Add the relevant NICs to the ASG.
- Create NSG rules using ASGs as source/destination.
- Associate the NSG with the appropriate subnet/NIC.
- Test communication between application tiers.

AZ-104 exam point: ASGs simplify logical grouping of application workloads in NSG rules.

Scenario 68: Evaluate effective NSG rules

Question / Business requirement: Evaluate effective NSG rules. Perform the configuration and verify the expected outcome.

Step-by-step solution

- Open a VM NIC → Effective security rules.
- Review inbound and outbound rules.
- Trace the relevant source, destination and port.
- Identify the first applicable rule by priority.
- Modify the conflicting NSG rule and retest.

AZ-104 exam point: Effective rules combine subnet and NIC NSGs.

Scenario 69: Deploy Azure Bastion

Question / Business requirement: Deploy Azure Bastion. Perform the configuration and verify the expected outcome.

Step-by-step solution

- Bastion → Create.
- Select the VNet and a dedicated supported subnet.
- Configure the required public IP.
- Create Bastion.
- Open a VM → Connect → Bastion and establish an RDP/SSH session.

AZ-104 exam point: Bastion provides browser-based management access without exposing VM RDP/SSH directly.

Scenario 70: Configure a service endpoint for Storage

Question / Business requirement: Configure a service endpoint for Storage. Perform the configuration and verify the expected outcome.

Step-by-step solution

- Open the target subnet.
- Enable the Microsoft.Storage service endpoint.
- Open the storage account networking settings.
- Allow the selected subnet.
- Test access from a VM in that subnet.

AZ-104 exam point: Service endpoints are subnet-level and retain the service's public endpoint.

Scenario 71: Configure a private endpoint for Azure SQL/Storage

Question / Business requirement: Configure a private endpoint for Azure SQL/Storage. Perform the configuration and verify the expected outcome.

Step-by-step solution

- Open the PaaS resource → Networking → Private endpoint.
- Select the target VNet/subnet.
- Configure private DNS integration.
- Create the endpoint.

- Verify DNS resolves the service name to the private IP from the VNet.

AZ-104 exam point: Private endpoint creates a private NIC in your VNet.

Scenario 72: Create an Azure DNS zone

Question / Business requirement: Create an Azure DNS zone. Perform the configuration and verify the expected outcome.

Step-by-step solution

- DNS zones → Create.
- Enter the domain name.
- Create the zone.
- Copy the assigned name servers.
- Delegate the domain at the registrar to the Azure DNS name servers.

AZ-104 exam point: Know the distinction between DNS zone hosting and domain registration.

Scenario 73: Create a DNS record

Question / Business requirement: Create a DNS record. Perform the configuration and verify the expected outcome.

Step-by-step solution

- Open the DNS zone.
- Select + Record set.
- Choose A, CNAME, MX, TXT, etc. as appropriate.
- Enter the record and TTL.
- Save and test name resolution.

AZ-104 exam point: Choose A for an IP address and CNAME for an alias to a hostname.

Scenario 74: Create an internal Load Balancer

Question / Business requirement: Create an internal Load Balancer. Perform the configuration and verify the expected outcome.

Step-by-step solution

- Load balancers → Create.
- Choose Internal.
- Configure frontend private IP.
- Add backend pool NICs/VMs.
- Configure health probe and load-balancing rule.

AZ-104 exam point: Internal Load Balancer exposes a private frontend.

Scenario 75: Create a public Load Balancer

Question / Business requirement: Create a public Load Balancer. Perform the configuration and verify the expected outcome.

Step-by-step solution

- Load balancers → Create.
- Choose Public.
- Configure public frontend IP.
- Add backend pool.
- Create health probe and load-balancing rule.
- Verify traffic distribution to healthy backends.

AZ-104 exam point: Health probes determine backend availability for load balancing.

Scenario 76: Troubleshoot an unhealthy Load Balancer backend

Question / Business requirement: Troubleshoot an unhealthy Load Balancer backend. Perform the configuration and verify the expected outcome.

Step-by-step solution

- Open the load balancer health probe.
- Verify probe protocol, port and path.
- Check NSG rules on the backend.
- Verify the application is listening on the probe port.
- Review effective routes and NIC configuration, then retest.

AZ-104 exam point: An unhealthy probe can remove a backend even when the VM itself is running.

Scenario 77: Configure a multi-tier network

Question / Business requirement: Configure a multi-tier network. Perform the configuration and verify the expected outcome.

Step-by-step solution

- Create web, application and database subnets.
- Associate appropriate NSGs/ASGs.
- Allow only web→app and app→database flows.
- Use private endpoints or service endpoints for PaaS dependencies.
- Test allowed and denied paths.

AZ-104 exam point: Think in terms of least privilege between tiers.

Scenario 78: Troubleshoot VM-to-VM connectivity

Question / Business requirement: Troubleshoot VM-to-VM connectivity. Perform the configuration and verify the expected outcome.

Step-by-step solution

- Check both VMs are in reachable VNets/subnets.
- Review NSGs and effective security rules.
- Review effective routes.
- Verify guest OS firewall and listening port.
- Use Network Watcher Connection troubleshoot to isolate the failure.

AZ-104 exam point: Separate Azure control-plane networking issues from guest OS issues.

Scenario 79: Configure NAT Gateway for outbound connectivity

Question / Business requirement: Configure NAT Gateway for outbound connectivity. Perform the configuration and verify the expected outcome.

Step-by-step solution

- Create a NAT Gateway.
- Associate a public IP/prefix.
- Associate the NAT Gateway with the target subnet.
- Deploy/test a VM in the subnet.
- Verify outbound traffic uses the NAT public IP.

AZ-104 exam point: NAT Gateway provides scalable outbound connectivity without exposing inbound access.

Scenario 80: Configure Network Watcher Connection Monitor

Question / Business requirement: Configure Network Watcher Connection Monitor. Perform the configuration and verify the expected outcome.

Step-by-step solution

- Open Network Watcher → Connection monitor.
- Select a source VM.
- Select destination hostname/IP and port.
- Configure monitoring frequency.
- Create the monitor and inspect latency/reachability results.

AZ-104 exam point: Connection Monitor is useful for continuous connectivity and latency diagnostics.

Domain 5 — Monitor and maintain Azure resources (10-15%)

Scenario 81: Create a VM metric alert

Question / Business requirement: Create a VM metric alert. Perform the configuration and verify the expected outcome.

Step-by-step solution

- Azure Monitor → Alerts → Create alert rule.
- Select the VM.
- Choose a metric such as Percentage CPU.
- Set the threshold and evaluation period.
- Attach an action group and create the rule.

AZ-104 exam point: Know metric alerts versus log-search alerts.

Scenario 82: Create an action group

Question / Business requirement: Create an action group. Perform the configuration and verify the expected outcome.

Step-by-step solution

- Azure Monitor → Alerts → Action groups → Create.
- Define notification channels such as email/SMS/webhook where supported.
- Add the required actions.
- Save the action group.
- Associate it with an alert rule.

AZ-104 exam point: Action groups define who/what gets notified or triggered.

Scenario 83: Create an alert processing rule

Question / Business requirement: Create an alert processing rule. Perform the configuration and verify the expected outcome.

Step-by-step solution

- Azure Monitor → Alerts → Alert processing rules.
- Select the target scope.
- Define conditions such as resource or severity.
- Choose actions such as suppress notifications during maintenance.
- Save and test the rule behavior.

AZ-104 exam point: Alert processing rules modify alert behavior without changing the underlying alert rule.

Scenario 84: Create a Log Analytics workspace

Question / Business requirement: Create a Log Analytics workspace. Perform the configuration and verify the expected outcome.

Step-by-step solution

- Log Analytics workspaces → Create.

- Select subscription/resource group/region.
- Create the workspace.
- Connect supported Azure resources or data collection rules.
- Open Logs and verify incoming data.

AZ-104 exam point: Log Analytics stores/query logs; Azure Monitor is the broader monitoring platform.

Scenario 85: Query Azure resource logs with KQL

Question / Business requirement: Query Azure resource logs with KQL. Perform the configuration and verify the expected outcome.

Step-by-step solution

- Open the Log Analytics workspace → Logs.
- Select an appropriate table.
- Write a KQL query filtering time/resource/severity.
- Run the query.
- Use summarize/project/order operations to analyze results.

AZ-104 exam point: Know the purpose of KQL and basic operators such as where, summarize, project and order by.

Scenario 86: Configure VM monitoring with Azure Monitor Insights

Question / Business requirement: Configure VM monitoring with Azure Monitor Insights. Perform the configuration and verify the expected outcome.

Step-by-step solution

- Open Azure Monitor → Virtual Machines/Insights.
- Select the VM.
- Enable monitoring and required data collection.
- Select or create a Log Analytics workspace.
- Review performance, dependencies and health information.

AZ-104 exam point: Insights provides workload-focused views built on Azure Monitor data.

Scenario 87: Monitor a storage account

Question / Business requirement: Monitor a storage account. Perform the configuration and verify the expected outcome.

Step-by-step solution

- Open the storage account → Monitoring/Insights or Metrics.
- Select metrics such as transactions, availability, latency or capacity.
- Adjust time range and aggregation.
- Pin/create a dashboard or alert based on the metric.
- Investigate anomalies using logs where enabled.

AZ-104 exam point: Know which metrics are appropriate for storage health and performance.

Scenario 88: Monitor network connectivity

Question / Business requirement: Monitor network connectivity. Perform the configuration and verify the expected outcome.

Step-by-step solution

- Open Network Watcher.
- Use Connection Monitor or connection troubleshoot.
- Select source and destination.
- Review reachability, latency and hop information.

- Correlate results with NSGs/routes/DNS.

AZ-104 exam point: Network Watcher is a key AZ-104 troubleshooting service.

Scenario 89: Create a Recovery Services vault

Question / Business requirement: Create a Recovery Services vault. Perform the configuration and verify the expected outcome.

Step-by-step solution

- Recovery Services vaults → Create.
- Choose subscription/resource group/region.
- Create the vault.
- Configure backup settings and storage redundancy as required.
- Use the vault for supported backup operations.

AZ-104 exam point: Recovery Services vault is used for supported backup and Site Recovery scenarios.

Scenario 90: Create an Azure Backup policy

Question / Business requirement: Create an Azure Backup policy. Perform the configuration and verify the expected outcome.

Step-by-step solution

- Open the backup vault/vault backup configuration.
- Choose workload type.
- Create a policy with schedule and retention.
- Associate the policy with the protected resource.
- Run/verify the first backup.

AZ-104 exam point: Know schedule versus retention settings.

Scenario 91: Back up an Azure VM

Question / Business requirement: Back up an Azure VM. Perform the configuration and verify the expected outcome.

Step-by-step solution

- Open the VM → Backup.
- Choose/create a vault.
- Select a backup policy.
- Enable backup.
- Run or wait for the first backup and verify the recovery point.

AZ-104 exam point: Backup creates recovery points independently of VM snapshots.

Scenario 92: Restore an Azure VM

Question / Business requirement: Restore an Azure VM. Perform the configuration and verify the expected outcome.

Step-by-step solution

- Open the vault → Backup items → Azure Virtual Machine.
- Select the protected VM.
- Choose a recovery point.
- Select the restore option and target configuration.
- Start restore and verify the resulting resources.

AZ-104 exam point: Know restore choices such as creating a new VM or restoring disks where supported.

Scenario 93: Configure backup alerts/reports

Question / Business requirement: Configure backup alerts/reports. Perform the configuration and verify the expected outcome.

Step-by-step solution

- Open the vault/Backup Center.
- Review backup jobs and health.
- Configure supported alerts/reporting.
- Set notification destinations.
- Test or verify alert delivery using backup job events.

AZ-104 exam point: Backup monitoring is part of maintaining recoverability.

Scenario 94: Configure Azure Site Recovery

Question / Business requirement: Configure Azure Site Recovery. Perform the configuration and verify the expected outcome.

Step-by-step solution

- Open Recovery Services vault/Site Recovery.
- Choose the Azure VM replication scenario.
- Configure source/target regions and replication settings.
- Enable replication.

Monitor initial synchronization and replication health.

AZ-104 exam point: Site Recovery is for disaster recovery/replication; Azure Backup is for point-in-time recovery.

Scenario 95: Perform a Site Recovery test failover

Question / Business requirement: Perform a Site Recovery test failover. Perform the configuration and verify the expected outcome.

Step-by-step solution

- Open the replicated item in Site Recovery.
- Choose Test failover.
- Select a suitable isolated network.
- Start the test failover.
- Validate the recovered VM/application and then clean up the test failover.

AZ-104 exam point: Test failover validates DR without disrupting production replication.

Scenario 96: Perform a planned/unplanned failover

Question / Business requirement: Perform a planned/unplanned failover. Perform the configuration and verify the expected outcome.

Step-by-step solution

- Open the replicated VM in Site Recovery.
- Choose the appropriate failover operation based on the scenario.
- Select the latest suitable recovery point.
- Start failover and monitor the job.
- Complete/commit or reprotect according to the recovery plan.

AZ-104 exam point: Know the difference between planned failover, unplanned failover, and test failover.

Scenario 97: Investigate a failed deployment with Activity Log

Question / Business requirement: Investigate a failed deployment with Activity Log. Perform the configuration and verify the expected outcome.

Step-by-step solution

- Open Monitor → Activity Log.
- Filter by subscription, resource group, resource and time.
- Open the failed operation.
- Review status, caller, operation and error details.
- Correct the root cause and retry.

AZ-104 exam point: Activity Log records control-plane operations; resource logs capture service/resource-level activity.

Scenario 98: Create a dashboard for operations

Question / Business requirement: Create a dashboard for operations. Perform the configuration and verify the expected outcome.

Step-by-step solution

- Azure portal → Dashboard → Create.
- Add metric charts, logs/query results, health information and resource tiles.
- Organize the dashboard by service/owner.
- Save and share according to organizational requirements.

AZ-104 exam point: Dashboards are for visualization; they do not replace alerts.

Scenario 99: Troubleshoot a VM that is running but unreachable

Question / Business requirement: Troubleshoot a VM that is running but unreachable. Perform the configuration and verify the expected outcome.

Step-by-step solution

- Check VM health and NIC configuration.
- Review public/private IP and NSG effective rules.
- Review effective routes.
- Use Network Watcher connection troubleshoot.
- Check guest OS firewall/service and verify the application is listening.

AZ-104 exam point: Always isolate the layer: Azure resource → network → guest OS → application.

Scenario 100: Create a combined operations review checklist

Question / Business requirement: Create a combined operations review checklist. Perform the configuration and verify the expected outcome.

Step-by-step solution

- Review Azure Service Health for relevant subscription issues.
- Review Advisor recommendations.
- Review Monitor alerts and Log Analytics trends.
- Review backup success/failure and recovery points.
- Review cost/budget status and governance compliance.
- Document remediation actions and owners.

AZ-104 exam point: AZ-104 administration is operational: monitoring, governance, recovery and troubleshooting all work together.

Final exam checklist

- Can I distinguish Azure RBAC roles, Entra directory roles, and resource scopes?
- Can I explain Policy versus RBAC versus resource locks?
- Can I choose the correct storage redundancy, access tier, SAS, private endpoint, or service endpoint?
- Can I administer managed disks, VM sizes, availability zones/sets, VMSS and VM moves?

- Can I deploy and troubleshoot App Service, ACR, ACI and Container Apps?
- Can I reason through NSG rules, ASGs, routes, DNS, Bastion, load balancers and private connectivity?
- Can I use Azure Monitor, Log Analytics/KQL, alerts and Network Watcher for troubleshooting?
- Can I configure Backup and Site Recovery and explain when each is appropriate?
- Can I read and modify ARM/Bicep deployments?
- Can I solve scenario questions by selecting the service that satisfies the requirement with the least complexity and appropriate security/cost?

Important source and scope note

This workbook is a practical study aid, not a dump of real exam questions. Microsoft states that its study guide describes skills that may be assessed and that related topics may also appear. Exam content can change, so always check the current Microsoft Learn AZ-104 study guide before booking the exam.

Official Microsoft Learn: <https://learn.microsoft.com/en-us/credentials/certifications/resources/study-guides/az-104>

Official certification page: <https://learn.microsoft.com/en-us/credentials/certifications/azure-administrator/>

AZ-104 Complete Expansion — Additional Scenarios

Purpose: These additional scenarios are appended to the original practical guide to close coverage gaps and deepen hands-on, command-line, troubleshooting, service-selection, and exam-decision practice. They are aligned to Microsoft's current AZ-104 Skills Measured objectives effective April 17, 2026.

Important: These are preparation scenarios, not leaked exam questions. The exam can test related topics in different wording.

Scenario 101: Use Azure Cloud Shell with Bash and PowerShell

Scenario / Requirement: Use Azure Cloud Shell with Bash and PowerShell

In simple words: You are an administrator who needs to run Azure commands without installing tools on your laptop.

Step-by-step solution:

- Open Azure Cloud Shell from the Azure portal.
- Choose Bash and verify the subscription with ``az account show``.
- Switch to PowerShell and run ``Get-AzContext``.
- Select the correct subscription before making changes.
- Run a harmless read-only command to confirm access.

AZ-104 exam point: Know that Cloud Shell provides both Bash/Azure CLI and PowerShell in a browser-based administration environment.

Scenario 102: Create a resource group using Azure CLI

Scenario / Requirement: Create a resource group using Azure CLI

In simple words: You need to create an Azure resource group from the command line.

Step-by-step solution:

- Open Cloud Shell Bash.
- Run ``az group create --name rg-az104-cli --location eastus``.
- Run ``az group show --name rg-az104-cli`` to verify it.
- Add tags with ``--tags Environment=Lab Owner=Admin`` if required.

AZ-104 exam point: Be comfortable translating a portal task into an Azure CLI command.

Scenario 103: Create a resource group using Azure PowerShell

Scenario / Requirement: Create a resource group using Azure PowerShell

In simple words: You need to perform the same administration task using PowerShell.

Step-by-step solution:

- Open Cloud Shell PowerShell.
- Run `New-AzResourceGroup -Name rg-az104-ps -Location EastUS``.
- Run `Get-AzResourceGroup -Name rg-az104-ps``.
- Confirm the subscription context before making changes.

AZ-104 exam point: Know the PowerShell equivalents of common Azure administration tasks.

Scenario 104: Find resources with Azure CLI and PowerShell

Scenario / Requirement: Find resources with Azure CLI and PowerShell

In simple words: You need to quickly identify resources by name, type, or resource group.

Step-by-step solution:

- Use `az resource list --resource-group <rg>``.
- Filter the output by resource type when needed.
- In PowerShell, use `Get-AzResource -ResourceGroupName <rg>``.
- Compare the resource IDs and locations before taking action.

AZ-104 exam point: Resource IDs, resource groups, and resource types are frequently used in administration and troubleshooting.

Scenario 105: Interpret an ARM template before deployment

Scenario / Requirement: Interpret an ARM template before deployment

In simple words: You are given an ARM JSON template and must determine what Azure resources it will create.

Step-by-step solution:

- Identify the `parameters``, `variables``, `resources``, and `outputs`` sections.
- Read the resource type and API version.
- Check dependencies and resource references.
- Determine which values must be supplied at deployment time.
- Do not deploy until you understand the resources and locations involved.

AZ-104 exam point: The current AZ-104 blueprint explicitly includes interpreting ARM templates.

Scenario 106: Modify an ARM template parameter

Scenario / Requirement: Modify an ARM template parameter

In simple words: You need to reuse an ARM template in another environment.

Step-by-step solution:

- Open the template parameters.
- Identify the parameter controlling location, VM size, or resource name.
- Change the parameter's allowed values or default value as required.
- Validate the JSON.
- Deploy to a test resource group and verify the resulting resources.

AZ-104 exam point: Know how to modify an existing ARM template rather than only deploy one.

Scenario 107: Modify an existing Bicep file

Scenario / Requirement: Modify an existing Bicep file

In simple words: You need to change an existing infrastructure definition without rebuilding it from scratch.

Step-by-step solution:

- Open the Bicep file.
- Identify parameters and resource declarations.

- Change the required property, such as SKU or location.
- Run a validation/what-if style check where appropriate.
- Deploy and verify the changed resource.

AZ-104 exam point: The current blueprint explicitly tests modifying Bicep files.

Scenario 108: Export an existing deployment as an ARM template

Scenario / Requirement: Export an existing deployment as an ARM template

In simple words: You have manually created Azure resources and want an infrastructure-as-code starting point.

Step-by-step solution:

- Open the resource group.
- Use the Export template capability.
- Review the generated template.
- Remove unnecessary or environment-specific values.
- Save and test the template in a separate resource group.

AZ-104 exam point: Know when exported templates are useful and understand that exported templates may need cleanup before reuse.

Scenario 109: Use ARM what-if before deployment

Scenario / Requirement: Use ARM what-if before deployment

In simple words: You want to know what a template deployment will change before applying it.

Step-by-step solution:

- Prepare the template and parameters.
- Run an ARM what-if operation.
- Review additions, modifications, and deletions.
- Correct unintended changes.
- Deploy only after the expected change set is understood.

AZ-104 exam point: What-if is useful for safe infrastructure changes and interpreting deployment impact.

Scenario 110: Change Microsoft Entra user properties

Scenario / Requirement: Change Microsoft Entra user properties

In simple words: You need to update a user's display name, job information, department, or account status.

Step-by-step solution:

- Open Microsoft Entra ID and Users.
- Select the user.
- Edit the required properties.
- Save the changes.
- Verify the updated account information.

AZ-104 exam point: Know the difference between user properties and Azure resource permissions.

Scenario 111: Assign a Microsoft Entra license to a user

Scenario / Requirement: Assign a Microsoft Entra license to a user

In simple words: A user needs a Microsoft cloud service license before they can use the required capability.

Step-by-step solution:

- Open Microsoft Entra ID > Licenses or the user's license settings.
- Select the required license.

- Assign it to the user or group.
- Verify the assignment.
- Check for any licensing prerequisites or conflicts.

AZ-104 exam point: Licensing in Entra ID is distinct from Azure RBAC permissions.

Scenario 112: Create a group and manage group properties

Scenario / Requirement: Create a group and manage group properties

In simple words: You need a reusable security boundary for assigning access to multiple users.

Step-by-step solution:

- Create a security group.
- Set the group name, description, and membership type.
- Add members.
- Verify membership.
- Use the group for an appropriate access assignment.

AZ-104 exam point: Understand security groups, membership, and how groups simplify access management.

Scenario 113: Determine whether a user is a guest or member

Scenario / Requirement: Determine whether a user is a guest or member

In simple words: You need to understand how an external collaborator is represented in the tenant.

Step-by-step solution:

- Open the user's Entra profile.
- Check the user type.
- Review the user's sign-in identity and invitation status.
- Confirm what resources the guest can access through assignments.

AZ-104 exam point: External users and their access are part of the current Entra objectives.

Scenario 114: Compare Owner, Contributor, Reader, and User Access Administrator

Scenario / Requirement: Compare Owner, Contributor, Reader, and User Access Administrator

In simple words: You must select the built-in Azure role that matches an administrator's required permissions.

Step-by-step solution:

- Identify whether the person must manage resources, view resources, or manage access.
- Compare the required permissions with the built-in roles.
- Choose the least-privileged role that satisfies the requirement.
- Assign it at the narrowest appropriate scope.

AZ-104 exam point: Be able to distinguish resource management permissions from permission-management permissions.

Scenario 115: Explain RBAC inheritance across scopes

Scenario / Requirement: Explain RBAC inheritance across scopes

In simple words: A user has access to a resource even though no role was assigned directly to that resource.

Step-by-step solution:

- Check the resource's direct assignments.
- Check the resource group assignments.

- Check subscription assignments.
- Check management-group assignments where applicable.
- Combine inherited permissions to determine effective access.

AZ-104 exam point: Azure RBAC permissions can flow down from broader scopes.

Scenario 116: Remove an unnecessary RBAC assignment

Scenario / Requirement: Remove an unnecessary RBAC assignment

In simple words: An administrator discovers a user has more access than required.

Step-by-step solution:

- Review the user's effective access.
- Identify the unnecessary role assignment and its scope.
- Remove only the excessive assignment.
- Recheck access.
- Confirm required access still works.

AZ-104 exam point: Least privilege and accurate interpretation of role assignments are important administration skills.

Scenario 117: Create a policy assignment with a remediation plan

Scenario / Requirement: Create a policy assignment with a remediation plan

In simple words: Resources already exist without a required configuration, and future resources must also comply.

Step-by-step solution:

- Choose an appropriate built-in policy.
- Assign it at the correct scope.
- Set parameters.
- Review existing non-compliant resources.
- Use remediation where the policy supports it.
- Recheck compliance.

AZ-104 exam point: Know the difference between preventing future non-compliance and remediating existing resources.

Scenario 118: Choose between a resource lock and Azure Policy

Scenario / Requirement: Choose between a resource lock and Azure Policy

In simple words: You need either to prevent accidental deletion or to enforce a configuration rule.

Step-by-step solution:

- If the requirement is deletion protection, consider a resource lock.
- If the requirement is a configuration/governance rule, consider Azure Policy.
- Apply the control at the appropriate scope.
- Test the expected behavior.

AZ-104 exam point: Locks and Policy solve different governance problems.

Scenario 119: Move a resource group resource while preserving dependencies

Scenario / Requirement: Move a resource group resource while preserving dependencies

In simple words: A resource must move to another resource group and you need to avoid breaking related resources.

Step-by-step solution:

- Review the resource's move-support requirements.
- Identify dependent resources.
- Check whether all required resources can move together.
- Perform the move.
- Validate resource IDs, dependencies, and application connectivity.

AZ-104 exam point: Resource moves can have dependency and support constraints.

Scenario 120: Compare management group, subscription, resource group, and resource scopes

Scenario / Requirement: Compare management group, subscription, resource group, and resource scopes

In simple words: You need to decide where to apply governance or permissions.

Step-by-step solution:

- Start at management group for broad organizational governance.
- Use subscription for subscription-wide controls.
- Use resource group for grouped workloads.
- Use resource scope for the narrowest targeted control.
- Choose the smallest scope that satisfies the requirement.

AZ-104 exam point: Scope selection is central to Azure governance and RBAC questions.

Scenario 121: Choose LRS, ZRS, GRS, GZRS, RA-GRS, or RA-GZRS

Scenario / Requirement: Choose LRS, ZRS, GRS, GZRS, RA-GRS, or RA-GZRS

In simple words: A workload has a specific durability and regional availability requirement.

Step-by-step solution:

- Determine whether the workload needs protection only within one datacenter, across availability zones, or across regions.
- Determine whether read access to the secondary region is required.
- Select the redundancy option that matches those requirements.
- Confirm service and region support.

AZ-104 exam point: Do not choose redundancy based only on the highest-looking acronym; map the requirement to the replication model.

Scenario 122: Configure a storage account with the correct performance and access tier

Scenario / Requirement: Configure a storage account with the correct performance and access tier

In simple words: You need to select storage settings based on workload behavior.

Step-by-step solution:

- Determine whether the workload needs standard or premium performance.
- For Blob data, determine whether Hot, Cool, Cold, or Archive behavior fits access frequency and retrieval needs.
- Configure the account and blob tier appropriately.
- Verify the resulting configuration.

AZ-104 exam point: Storage performance and Blob access tier are different decisions.

Scenario 123: Recover a deleted blob using blob soft delete

Scenario / Requirement: Recover a deleted blob using blob soft delete

In simple words: A user accidentally deletes a blob and the retention period is still active.

Step-by-step solution:

- Open the storage account's data protection settings.
- Confirm blob soft delete is enabled and retention is sufficient.
- Open the container and locate deleted blobs.
- Undelete the required blob.
- Verify that the object is available again.

AZ-104 exam point: Understand soft delete as protection against accidental deletion.

Scenario 124: Recover a deleted container

Scenario / Requirement: Recover a deleted container

In simple words: An entire Blob container was accidentally deleted.

Step-by-step solution:

- Verify container soft delete is enabled.
- Check the retention window.
- Use the storage account recovery capability to restore the container if available.
- Verify the container and its data.

AZ-104 exam point: Container soft delete is distinct from blob soft delete.

Scenario 125: Use blob versioning to restore an earlier version

Scenario / Requirement: Use blob versioning to restore an earlier version

In simple words: A file was overwritten and the previous version must be recovered.

Step-by-step solution:

- Verify blob versioning is enabled.
- Open the blob's versions.
- Identify the required earlier version.
- Promote/copy the required version as appropriate.
- Verify the restored content.

AZ-104 exam point: Versioning protects against overwrites and creates recoverable object versions.

Scenario 126: Create a lifecycle rule based on blob age

Scenario / Requirement: Create a lifecycle rule based on blob age

In simple words: Old data should automatically move to a cheaper tier and eventually be deleted.

Step-by-step solution:

- Open lifecycle management.
- Create a rule for the target container or blob prefix.
- Set a move-to-tier condition based on last modification age.
- Optionally set a delete condition.
- Review the rule and save it.

AZ-104 exam point: Lifecycle management automates storage-cost and retention behavior.

Scenario 127: Generate a SAS with minimum required permissions

Scenario / Requirement: Generate a SAS with minimum required permissions

In simple words: An application needs temporary read-only access to one Blob object.

Step-by-step solution:

- Choose the narrowest resource scope possible.
- Grant only read permission.
- Set a short start/expiry period.
- Restrict the protocol and IP range when appropriate.
- Generate and test the SAS.

AZ-104 exam point: SAS questions often test least privilege, scope, permissions, and expiry.

Scenario 128: Revoke access after a SAS compromise

Scenario / Requirement: Revoke access after a SAS compromise

In simple words: A SAS token has been exposed and must no longer be trusted.

Step-by-step solution:

- Identify how the SAS was issued.
- If based on a stored access policy, change or revoke the policy as appropriate.
- If account-level credentials were exposed, rotate the relevant keys.
- Issue a replacement SAS with narrower permissions.
- Verify the old access no longer works where applicable.

AZ-104 exam point: Know that the revocation method depends on how the SAS was created.

Scenario 129: Use a stored access policy for centrally managed SAS

Scenario / Requirement: Use a stored access policy for centrally managed SAS

In simple words: Multiple SAS tokens need a centrally controlled permission and expiration policy.

Step-by-step solution:

- Create the stored access policy on the container.
- Set permissions and expiry.
- Generate SAS tokens that reference the policy.
- When access must be changed, update or revoke the policy.
- Test the resulting access.

AZ-104 exam point: Stored access policies can simplify centralized SAS management.

Scenario 130: Rotate storage account access keys

Scenario / Requirement: Rotate storage account access keys

In simple words: A storage account key may have been exposed and applications must continue working during rotation.

Step-by-step solution:

- Identify which key applications currently use.
- Regenerate the unused key first.
- Update dependent applications to the new key.
- Regenerate the old key.

Monitor applications for authentication failures.

AZ-104 exam point: Key rotation is safer when applications are moved from one key to the other before regenerating the active key.

Scenario 131: Restrict storage access to selected networks

Scenario / Requirement: Restrict storage access to selected networks

In simple words: Only approved VNets or IP ranges should access the storage account.

Step-by-step solution:

- Open storage networking settings.
- Choose the appropriate network access model.
- Add allowed virtual networks or IP ranges.
- Configure the required endpoint type.
- Test allowed and denied clients.

AZ-104 exam point: Be able to distinguish public firewall restrictions from private endpoint-based access.

Scenario 132: Choose Service Endpoint versus Private Endpoint

Scenario / Requirement: Choose Service Endpoint versus Private Endpoint

In simple words: A storage service must be reachable from a VNet, and the security requirement determines the design.

Step-by-step solution:

- If the requirement is VNet-based service access while the PaaS service remains on its public endpoint, evaluate a service endpoint.
- If the requirement is a private IP in the VNet and private connectivity to the PaaS resource, evaluate a private endpoint.
- Check DNS and network implications.
- Test connectivity.

AZ-104 exam point: This distinction is a common Azure networking/storage decision point.

Scenario 133: Configure Azure Files identity-based access

Scenario / Requirement: Configure Azure Files identity-based access

In simple words: Users should access an Azure file share using identity-based authentication rather than storage keys.

Step-by-step solution:

- Configure the required identity-based authentication model.
- Prepare the relevant identities/groups and permissions.
- Create the file share.
- Assign share-level and file/directory permissions as required.
- Test access using a user account.

AZ-104 exam point: Identity-based Azure Files access is explicitly listed in the current exam objectives.

Scenario 134: Create an Azure Files snapshot and restore a file

Scenario / Requirement: Create an Azure Files snapshot and restore a file

In simple words: A file in an Azure file share was changed and you need an earlier copy.

Step-by-step solution:

- Open the file share.
- Create or locate a share snapshot.
- Browse the snapshot contents.
- Copy the required file back to the live share.
- Verify the restored file.

AZ-104 exam point: Know what an Azure Files snapshot protects and how it differs from blob versions.

Scenario 135: Use AzCopy to copy a local directory to Blob Storage

Scenario / Requirement: Use AzCopy to copy a local directory to Blob Storage

In simple words: You need a fast command-line transfer of many files into a Blob container.

Step-by-step solution:

- Authenticate to Azure using the supported AzCopy method.
- Identify the source directory and destination container.
- Run the appropriate ``azcopy copy`` command.
- Review transfer output.
- Verify the uploaded blobs.

AZ-104 exam point: Know when AzCopy is appropriate for command-line data movement.

Scenario 136: Use Storage Explorer to inspect and move data

Scenario / Requirement: Use Storage Explorer to inspect and move data

In simple words: You need a graphical tool to browse storage data and perform administration tasks.

Step-by-step solution:

- Open Azure Storage Explorer.
- Sign in or connect using the required method.
- Navigate to the storage account and target service.
- Inspect containers, blobs, or file shares.
- Perform a small test upload/download.

AZ-104 exam point: Storage Explorer is explicitly included in the current storage objectives.

Scenario 137: Choose a VM size based on workload requirements

Scenario / Requirement: Choose a VM size based on workload requirements

In simple words: A VM needs a certain CPU, memory, disk, or network capability.

Step-by-step solution:

- Identify the workload bottleneck.
- Compare VM families and sizes.
- Check regional availability and quotas.
- Choose a size that meets the requirement.
- Resize the VM if necessary and verify performance.

AZ-104 exam point: VM size selection is a requirements-matching problem, not simply choosing the largest VM.

Scenario 138: Attach and initialize a data disk

Scenario / Requirement: Attach and initialize a data disk

In simple words: A Windows VM needs additional persistent storage for application data.

Step-by-step solution:

- Create or attach a managed disk.
- Start the VM and open Disk Management.
- Bring the disk online if necessary.
- Initialize and format the disk.
- Assign a drive letter and verify persistence.

AZ-104 exam point: Understand OS disk versus data disk and managed disk administration.

Scenario 139: Create a VM with multiple data disks

Scenario / Requirement: Create a VM with multiple data disks

In simple words: A workload requires separate disks for application data and logs.

Step-by-step solution:

- Create the VM.
- Add the required managed disks.
- Choose appropriate disk SKUs.
- Initialize the disks inside the guest OS.
- Place the application and logs on the intended disks.

AZ-104 exam point: Disk layout affects performance, management, and workload separation.

Scenario 140: Move a VM to another region using a supported migration approach

Scenario / Requirement: Move a VM to another region using a supported migration approach

In simple words: A VM must run in a different Azure region.

Step-by-step solution:

- Check whether the resource type and configuration support the chosen move method.
- Identify dependencies such as NICs, disks, public IPs, and NSGs.
- Use an appropriate Azure migration/copy/deployment approach.
- Deploy the target resources in the destination region.
- Validate the workload before retiring the source.

AZ-104 exam point: A region move is not the same operation as moving a VM between resource groups.

Scenario 141: Compare Availability Zones and Availability Sets

Scenario / Requirement: Compare Availability Zones and Availability Sets

In simple words: You need resilience against hardware/rack/datacenter-level failures.

Step-by-step solution:

- Identify the failure domain the workload must tolerate.
- For zone-level separation within a region, evaluate Availability Zones.
- For fault/update domain separation where zones are unavailable or not required, evaluate Availability Sets.
- Deploy redundant VMs according to the chosen model.

AZ-104 exam point: Know what each availability model protects against.

Scenario 142: Configure VM Scale Set instance count

Scenario / Requirement: Configure VM Scale Set instance count

In simple words: A web workload needs multiple VM instances and automatic scaling.

Step-by-step solution:

- Create a VM Scale Set.
- Set the initial instance count.
- Configure autoscale rules based on a metric.
- Generate load or simulate metric changes.
- Verify instances are added or removed.

AZ-104 exam point: Understand scale-out versus scale-in and the relationship between minimum, default, and maximum instances.

Scenario 143: Configure VM Scale Set upgrade behavior

Scenario / Requirement: Configure VM Scale Set upgrade behavior

In simple words: You need to control how model changes are applied to VM instances.

Step-by-step solution:

- Open the VM Scale Set upgrade policy.
- Understand the available upgrade behavior for the scenario.
- Apply a controlled change.
- Observe instance updates.
- Verify that the desired model is applied consistently.

AZ-104 exam point: VMSS administration can involve both scale behavior and model/instance update behavior.

Scenario 144: Enable VM boot diagnostics and investigate startup problems

Scenario / Requirement: Enable VM boot diagnostics and investigate startup problems

In simple words: A VM is failing to boot and you need console or boot information.

Step-by-step solution:

- Enable boot diagnostics.
- Restart the VM if necessary.
- Review boot logs/screenshots where available.
- Correlate findings with Activity Log and guest/extension information.
- Apply the corrective action and retest.

AZ-104 exam point: Boot diagnostics is a practical troubleshooting tool for VM startup issues.

Scenario 145: Deploy and troubleshoot a VM extension

Scenario / Requirement: Deploy and troubleshoot a VM extension

In simple words: You need to install an agent or perform configuration on a VM using an extension.

Step-by-step solution:

- Choose the appropriate VM extension.
- Provide the required settings.
- Deploy the extension.
- Check extension provisioning status.
- Review logs if it fails.
- Correct configuration and retry.

AZ-104 exam point: Extensions are managed resources attached to VMs and can fail independently of the VM itself.

Scenario 146: Configure App Service plan scaling

Scenario / Requirement: Configure App Service plan scaling

In simple words: A web application needs more capacity as traffic increases.

Step-by-step solution:

- Determine whether the requirement is scale up or scale out.
- For scale up, choose a different App Service plan tier/size.
- For scale out, increase instance count or configure autoscale where supported.
- Apply the change.

Monitor application performance.

AZ-104 exam point: Know the difference between changing the plan's capabilities and adding instances.

Scenario 147: Configure App Service application settings and connection strings

Scenario / Requirement: Configure App Service application settings and connection strings

In simple words: An application needs environment-specific configuration without changing application code.

Step-by-step solution:

- Open Configuration for the App Service.
- Add the required application settings or connection strings.
- Save the configuration.
- Restart/reload if required.
- Verify the application reads the values correctly.

AZ-104 exam point: Configuration should be managed separately from application code when appropriate.

Scenario 148: Configure App Service networking

Scenario / Requirement: Configure App Service networking

In simple words: An App Service must communicate with resources in a VNet or control inbound exposure.

Step-by-step solution:

- Identify whether the requirement is outbound VNet integration or inbound private access.
- Configure the supported App Service networking feature.
- Verify subnet and network requirements.
- Test connectivity to the target resource.
- Check DNS if private connectivity is involved.

AZ-104 exam point: The current blueprint explicitly includes App Service networking settings.

Scenario 149: Configure App Service deployment slots

Scenario / Requirement: Configure App Service deployment slots

In simple words: A new version must be tested before production release.

Step-by-step solution:

- Create a staging slot.
- Deploy the new version to the staging slot.
- Test the application.
- Review slot settings and sticky configuration.
- Swap staging with production.

Monitor the production slot after the swap.

AZ-104 exam point: Deployment slots support controlled releases and low-downtime deployment patterns.

Scenario 150: Configure an App Service custom domain and TLS certificate

Scenario / Requirement: Configure an App Service custom domain and TLS certificate

In simple words: Users must access the web app through a business domain over HTTPS.

Step-by-step solution:

- Add the custom hostname.
- Create the required DNS record.
- Verify domain ownership.

- Configure the certificate/TLS binding.
- Test HTTPS access and certificate validity.

AZ-104 exam point: Custom DNS mapping and TLS are separate configuration steps.

Scenario 151: Configure an App Service backup

Scenario / Requirement: Configure an App Service backup

In simple words: You need periodic recoverable copies of the web app.

Step-by-step solution:

- Confirm the App Service plan and storage requirements.
- Create/configure the backup destination.
- Set the backup schedule and retention as supported.
- Run or wait for a backup.
- Test restoration in a safe environment.

AZ-104 exam point: Know the configuration and prerequisites for App Service backup.

Scenario 152: Deploy a container image from ACR to a container workload

Scenario / Requirement: Deploy a container image from ACR to a container workload

In simple words: A container image is stored privately in Azure Container Registry and must be run by an Azure container service.

Step-by-step solution:

- Create or identify the ACR.
- Push/tag the image.
- Configure the target container service.
- Provide the required ACR authentication/managed identity configuration.
- Deploy the image.
- Verify the running container.

AZ-104 exam point: Understand the relationship between image registry and container runtime.

Scenario 153: Calculate a subnet address range from a CIDR requirement

Scenario / Requirement: Calculate a subnet address range from a CIDR requirement

In simple words: You need to create a subnet large enough for a workload while avoiding overlap.

Step-by-step solution:

- Identify the VNet address space.
- Determine the required number of usable addresses.
- Choose an appropriate subnet prefix.
- Verify that the subnet does not overlap existing subnets.
- Create the subnet.

AZ-104 exam point: CIDR/subnet planning is foundational to VNet administration.

Scenario 154: Create a VNet with non-overlapping subnets

Scenario / Requirement: Create a VNet with non-overlapping subnets

In simple words: A three-tier application needs separate web, application, and database subnets.

Step-by-step solution:

- Choose a VNet address space.
- Create non-overlapping subnet prefixes.
- Create the VNet.
- Create each subnet.
- Associate security controls as required.
- Verify address ranges.

AZ-104 exam point: Subnet design and non-overlapping address spaces are essential for Azure networking.

Scenario 155: Configure VNet peering and verify bidirectional connectivity

Scenario / Requirement: Configure VNet peering and verify bidirectional connectivity

In simple words: Two VNets need private connectivity.

Step-by-step solution:

- Create the peering from VNet A to VNet B.
- Create the reciprocal peering.
- Enable required peering settings.
- Check peering status on both sides.
- Test connectivity between resources.

AZ-104 exam point: VNet peering is directional configuration on each VNet and both sides must be correctly configured.

Scenario 156: Troubleshoot overlapping VNet address spaces

Scenario / Requirement: Troubleshoot overlapping VNet address spaces

In simple words: Two VNets cannot be peered because their address ranges overlap.

Step-by-step solution:

- Inspect both VNet address spaces.
- Identify the overlapping CIDR ranges.
- Determine which network can be renumbered.
- Redesign the affected subnet/VNet range.
- Reconfigure dependent resources and retry peering.

AZ-104 exam point: Overlapping address spaces are a fundamental connectivity constraint.

Scenario 157: Create a UDR to send traffic to an NVA

Scenario / Requirement: Create a UDR to send traffic to an NVA

In simple words: Traffic to a specific destination must pass through a network virtual appliance.

Step-by-step solution:

- Create or identify the route table.
- Add a route for the target prefix.
- Set the next hop type to Virtual Appliance.
- Enter the NVA private IP.
- Associate the route table with the subnet.
- Verify effective routes.

AZ-104 exam point: Know route-table association and next-hop concepts.

Scenario 158: Diagnose an NSG rule that blocks traffic

Scenario / Requirement: Diagnose an NSG rule that blocks traffic

In simple words: A VM cannot receive traffic even though the service is running.

Step-by-step solution:

- Check NSGs associated with the subnet and NIC.
- Review inbound rules by priority.
- Identify the first applicable allow/deny behavior.
- Check source, destination, protocol, and port.
- Modify the smallest necessary rule.
- Retest connectivity.

AZ-104 exam point: Effective security rules and priority ordering are key troubleshooting skills.

Scenario 159: Use an Application Security Group in an NSG rule

Scenario / Requirement: Use an Application Security Group in an NSG rule

In simple words: You want to allow web-tier VMs to communicate with application-tier VMs without hard-coding IP addresses.

Step-by-step solution:

- Create ASGs for the relevant tiers.
- Associate the VM NICs with the correct ASGs.
- Create an NSG rule using the source and destination ASGs.
- Set the required port/protocol and priority.
- Test connectivity.

AZ-104 exam point: ASGs simplify role-based network rules.

Scenario 160: Deploy Azure Bastion for private VM administration

Scenario / Requirement: Deploy Azure Bastion for private VM administration

In simple words: Administrators need browser-based RDP/SSH without public IPs on the VMs.

Step-by-step solution:

- Create or identify the VNet.
- Create the required AzureBastionSubnet.
- Deploy Azure Bastion.
- Ensure the target VM is reachable through the VNet.
- Connect through the Azure portal.
- Verify that direct public management exposure is not required.

AZ-104 exam point: Bastion provides managed administrative access without exposing VM RDP/SSH ports directly.

Scenario 161: Create a private endpoint and configure DNS

Scenario / Requirement: Create a private endpoint and configure DNS

In simple words: A PaaS service must be accessible privately from a VNet.

Step-by-step solution:

- Create the private endpoint in the target subnet.
- Select the PaaS resource.
- Configure the appropriate private DNS integration.
- Verify the private IP assigned to the endpoint.
- Resolve the service hostname from a VNet resource.

- Test connectivity.

AZ-104 exam point: Private endpoint connectivity commonly depends on correct DNS resolution.

Scenario 162: Create a service endpoint for a PaaS service

Scenario / Requirement: Create a service endpoint for a PaaS service

In simple words: A subnet should be allowed to access a supported Azure service using the service endpoint model.

Step-by-step solution:

- Enable the required service endpoint on the subnet.
- Configure the target PaaS resource's network rules.
- Add the appropriate VNet/subnet access.
- Test access from an allowed subnet.
- Test from a disallowed network if applicable.

AZ-104 exam point: Service endpoints and private endpoints provide different connectivity/security models.

Scenario 163: Create an Azure DNS zone and A record

Scenario / Requirement: Create an Azure DNS zone and A record

In simple words: A domain name must resolve to an Azure-hosted service.

Step-by-step solution:

- Create the DNS zone.
- Create an A record with the required IP address.
- Identify the name servers assigned to the zone.
- Delegate the domain at the registrar if required.
- Use DNS tools to verify resolution.

AZ-104 exam point: Know zones, records, delegation, and name resolution.

Scenario 164: Troubleshoot an Azure DNS resolution failure

Scenario / Requirement: Troubleshoot an Azure DNS resolution failure

In simple words: A hostname exists in Azure DNS but clients cannot resolve it.

Step-by-step solution:

- Confirm the record exists.
- Check the authoritative name servers.
- Verify domain delegation.
- Check TTL/caching.
- Use `nslookup` or `dig` to test resolution.
- Correct the DNS configuration and retest.

AZ-104 exam point: DNS troubleshooting is about determining which layer—record, delegation, resolver, or cache—is failing.

Scenario 165: Configure an internal Load Balancer

Scenario / Requirement: Configure an internal Load Balancer

In simple words: Multiple backend VMs must expose an application through one private IP.

Step-by-step solution:

- Create a frontend private IP.
- Create a backend pool.
- Configure a health probe.

- Create a load-balancing rule.
- Add backend VMs.
- Test the frontend IP from the VNet.

AZ-104 exam point: Internal Load Balancers provide private load distribution.

Scenario 166: Configure a public Load Balancer

Scenario / Requirement: Configure a public Load Balancer

In simple words: Internet clients must reach a highly available service hosted on multiple VMs.

Step-by-step solution:

- Create a public frontend IP.
- Create a backend pool.
- Create a health probe.
- Create a load-balancing rule.
- Add backend VMs.
- Test from an external client.

AZ-104 exam point: Understand the relationship among frontend IP, backend pool, health probe, and rule.

Scenario 167: Troubleshoot an unhealthy Load Balancer backend

Scenario / Requirement: Troubleshoot an unhealthy Load Balancer backend

In simple words: A VM is in the backend pool but receives no traffic.

Step-by-step solution:

- Check the health probe protocol, port, and path where applicable.
- Verify the service is listening on the probe port.
- Check NSG rules.
- Check the VM guest firewall.
- Review load-balancer rule configuration.
- Retest backend health.

AZ-104 exam point: A healthy backend requires both a working service and reachable health probe.

Scenario 168: Use Network Watcher IP Flow Verify

Scenario / Requirement: Use Network Watcher IP Flow Verify

In simple words: You need to determine whether an NSG allows or denies a particular flow.

Step-by-step solution:

- Open Network Watcher.
- Select IP Flow Verify.
- Specify source, destination, protocol, and port.
- Run the test.
- Identify the rule causing the result.
- Correct the network rule if appropriate.

AZ-104 exam point: Network Watcher is an important practical troubleshooting service.

Scenario 169: Use Network Watcher Next Hop

Scenario / Requirement: Use Network Watcher Next Hop

In simple words: Traffic from a VM is not reaching its destination and you need to know the route Azure selects.

Step-by-step solution:

- Open Network Watcher Next Hop.
- Select the VM/NIC and destination IP.
- Run the check.
- Review the selected next hop and route.
- Inspect route tables or peering if the result is unexpected.

AZ-104 exam point: Next Hop helps troubleshoot routing decisions.

Scenario 170: Compare Load Balancer, Application Gateway, and Front Door

Scenario / Requirement: Compare Load Balancer, Application Gateway, and Front Door

In simple words: A web application needs a specific form of traffic distribution.

Step-by-step solution:

- For L4 TCP/UDP distribution, evaluate Load Balancer.
- For regional HTTP/S application routing and Layer 7 features, evaluate Application Gateway.
- For global web delivery and edge routing, evaluate Front Door.
- Choose based on protocol, scope, and required features.

AZ-104 exam point: Service-selection questions require matching the service to the traffic and deployment requirement.

Scenario 171: Troubleshoot VM-to-VM connectivity systematically

Scenario / Requirement: Troubleshoot VM-to-VM connectivity systematically

In simple words: Two VMs in Azure cannot communicate.

Step-by-step solution:

- Check that both VMs are running.
- Check VNet/subnet and address ranges.
- Check NSGs on NICs and subnets.
- Check effective routes.
- Check guest OS firewall.
- Check that the destination service is listening.
- Use Network Watcher tools to isolate the failure.

AZ-104 exam point: Use a layered troubleshooting approach rather than changing random settings.

Scenario 172: Create a Log Analytics workspace and connect logs

Scenario / Requirement: Create a Log Analytics workspace and connect logs

In simple words: You need centralized queryable logs for Azure resources.

Step-by-step solution:

- Create the Log Analytics workspace.
- Open the target resource's diagnostic settings.
- Select the workspace as the destination.
- Choose relevant log categories and metrics.
- Generate activity and wait for ingestion.
- Query the data.

AZ-104 exam point: Diagnostic settings determine what resource logs are sent to destinations such as Log Analytics.

Scenario 173: Write a KQL query to find failed operations

Scenario / Requirement: Write a KQL query to find failed operations

In simple words: You need to investigate failed administrative or resource operations.

Step-by-step solution:

- Open Logs in the Log Analytics workspace.
- Identify the relevant table.
- Filter records for failures or errors.
- Add a time filter.
- Project useful columns such as time, operation, resource, and status.
- Sort newest first.

AZ-104 exam point: Know basic KQL filtering, projection, sorting, and time scoping.

Scenario 174: Configure a diagnostic setting for a storage account

Scenario / Requirement: Configure a diagnostic setting for a storage account

In simple words: Storage operations need to be sent to a centralized logging destination.

Step-by-step solution:

- Open the storage account's diagnostic settings.
- Choose the appropriate resource log categories.
- Select Log Analytics or another supported destination.
- Save the setting.
- Generate activity and query the resulting logs.

AZ-104 exam point: Monitoring requires both enabling data collection and knowing where that data is stored.

Scenario 175: Create an Azure Monitor alert with an action group

Scenario / Requirement: Create an Azure Monitor alert with an action group

In simple words: You want a notification when a VM's CPU remains above a threshold.

Step-by-step solution:

- Create or select an action group.
- Create an alert rule for the VM.
- Choose the CPU metric.
- Set the threshold and evaluation window.
- Attach the action group.
- Save and test the alert.

AZ-104 exam point: Know the separation between the alert rule and its notification/action configuration.

Scenario 176: Use an alert processing rule to suppress notifications

Scenario / Requirement: Use an alert processing rule to suppress notifications

In simple words: Maintenance is scheduled and you do not want routine alerts to notify administrators during the window.

Step-by-step solution:

- Create an alert processing rule.
- Define the target scope.
- Choose the suppression/schedule behavior.
- Set the maintenance time window.
- Verify that alerts are still generated but notifications are processed according to the rule.

AZ-104 exam point: Alert processing rules control how alerts are handled after they are generated.

Scenario 177: Interpret VM metrics versus logs

Scenario / Requirement: Interpret VM metrics versus logs

In simple words: You need to decide whether a performance problem is best investigated through metrics or logs.

Step-by-step solution:

- Use metrics for time-series measurements such as CPU percentage.
- Use logs for detailed event/query analysis.
- Select an appropriate time range.
- Correlate metrics and logs to determine the likely cause.
- Create an alert if ongoing monitoring is required.

AZ-104 exam point: Know when metrics and logs answer different operational questions.

Scenario 178: Use VM Insights to investigate performance

Scenario / Requirement: Use VM Insights to investigate performance

In simple words: A VM is experiencing performance issues and you need deeper monitoring data.

Step-by-step solution:

- Enable the required VM monitoring/Insights capability.
- Open VM Insights.
- Review CPU, memory, disk, and network information available.
- Correlate with time ranges and logs.
- Use the findings to identify the bottleneck.

AZ-104 exam point: VM Insights is a monitoring experience built on Azure Monitor capabilities.

Scenario 179: Use Storage Insights to investigate a storage issue

Scenario / Requirement: Use Storage Insights to investigate a storage issue

In simple words: A storage account is showing unusual traffic or performance behavior.

Step-by-step solution:

- Open the storage account's monitoring/Insights view.
- Review transactions, availability, latency, and capacity where available.
- Select the affected time window.
- Correlate with activity and resource logs.
- Take corrective action.

AZ-104 exam point: Storage monitoring helps distinguish capacity, transaction, availability, and latency issues.

Scenario 180: Use Network Insights to investigate connectivity

Scenario / Requirement: Use Network Insights to investigate connectivity

In simple words: A network resource has connectivity or performance problems.

Step-by-step solution:

- Open the relevant network monitoring/Insights experience.
- Review topology and connectivity information.
- Check relevant metrics and diagnostic data.
- Use Network Watcher tools for targeted tests.
- Correct the identified routing/security issue.

AZ-104 exam point: Azure Monitor and Network Watcher complement each other in network troubleshooting.

Scenario 181: Create an Azure Backup vault

Scenario / Requirement: Create an Azure Backup vault

In simple words: You need to use the newer vault model for a supported backup workload.

Step-by-step solution:

- Open Backup Center or the appropriate Azure backup experience.
- Create an Azure Backup vault in the target region.
- Configure appropriate redundancy/settings.
- Select a supported workload and create protection.
- Verify the protected item appears in the vault.

AZ-104 exam point: The current AZ-104 blueprint explicitly includes creating an Azure Backup vault in addition to Recovery Services vault.

Scenario 182: Choose Recovery Services vault versus Azure Backup vault

Scenario / Requirement: Choose Recovery Services vault versus Azure Backup vault

In simple words: You are asked which vault type should be used for a backup requirement.

Step-by-step solution:

- Identify the workload and backup service being used.
- Check whether the scenario calls for Recovery Services vault or Azure Backup vault.
- Create the appropriate vault in the correct region.
- Configure protection and verify the protected item.

AZ-104 exam point: Know that Microsoft now distinguishes these vault concepts in the current exam objectives.

Scenario 183: Configure an Azure Backup policy

Scenario / Requirement: Configure an Azure Backup policy

In simple words: A VM needs scheduled backups with a defined retention period.

Step-by-step solution:

- Create or select the appropriate vault.
- Create the backup policy.
- Set the schedule.
- Set retention according to the requirement.
- Apply the policy to the VM.
- Verify the protected item and next backup.

AZ-104 exam point: Backup-policy questions often test schedule and retention settings.

Scenario 184: Perform an on-demand VM backup

Scenario / Requirement: Perform an on-demand VM backup

In simple words: A backup is required immediately before a risky change.

Step-by-step solution:

- Open the protected VM in the backup vault.
- Start an on-demand backup.
- Choose the retention option if prompted.

Monitor the backup job.

- Verify the recovery point exists.

AZ-104 exam point: Understand the difference between scheduled backups and on-demand backup operations.

Scenario 185: Restore an Azure VM to a new location

Scenario / Requirement: Restore an Azure VM to a new location

In simple words: A VM must be recovered without overwriting the original workload.

Step-by-step solution:

- Open the vault's backup item.
- Choose a recovery point.
- Select restore options.
- Choose a new VM/resource group/network where supported.
- Start the restore.
- Validate the restored VM.

AZ-104 exam point: Know restore-target choices and why restoring to a new location can be safer for investigation.

Scenario 186: Configure backup reports and alerts

Scenario / Requirement: Configure backup reports and alerts

In simple words: Operations teams need visibility into failed backups and backup health.

Step-by-step solution:

- Configure the required monitoring/reporting integration.
- Select the relevant backup scope.
- Configure alerts for failed or unhealthy operations.
- Set notification recipients through an action mechanism.
- Test the monitoring workflow.

AZ-104 exam point: Backup monitoring is part of the current exam objectives, not just performing backups.

Scenario 187: Configure Site Recovery for an Azure VM

Scenario / Requirement: Configure Site Recovery for an Azure VM

In simple words: A workload needs disaster recovery to another Azure region.

Step-by-step solution:

- Create or identify the required Site Recovery components.
- Choose the source and target regions.
- Enable replication for the VM.
- Configure target resource settings.

Monitor replication health.

- Document the recovery plan.

AZ-104 exam point: Know the purpose of Site Recovery and the distinction between replication and backup.

Scenario 188: Perform a Site Recovery test failover

Scenario / Requirement: Perform a Site Recovery test failover

In simple words: You want to validate disaster recovery without disrupting production.

Step-by-step solution:

- Open the replicated item.
- Start a test failover using an appropriate recovery point.
- Select an isolated/test network where applicable.
- Validate the recovered VM and application.

- Clean up the test failover according to the procedure.

AZ-104 exam point: A test failover validates DR readiness without performing the production failover.

Scenario 189: Perform a planned Site Recovery failover

Scenario / Requirement: Perform a planned Site Recovery failover

In simple words: The primary region is available but a controlled disaster-recovery switch is required.

Step-by-step solution:

- Confirm replication health.
- Prepare the target environment.
- Initiate planned failover.
- Validate target workloads.
- Complete the failover procedure.
- Reverse-protect/reprotect as required after stabilization.

AZ-104 exam point: Planned and unplanned failover have different operational contexts.

Scenario 190: Investigate a failed Azure deployment using Activity Log

Scenario / Requirement: Investigate a failed Azure deployment using Activity Log

In simple words: A deployment failed and you need to determine why.

Step-by-step solution:

- Open Activity Log at the relevant scope.
- Filter by time and failed operations.
- Open the failed operation.
- Read the error code and message.
- Check the affected resource and policy/RBAC constraints.
- Correct the issue and redeploy.

AZ-104 exam point: Activity Log is a first-line source for control-plane troubleshooting.

Scenario 191: Build an operations dashboard

Scenario / Requirement: Build an operations dashboard

In simple words: An administrator wants a single view of important VM, storage, and alert information.

Step-by-step solution:

- Create or open an Azure dashboard.
- Add relevant monitoring tiles/charts.
- Add key metrics and resource health information.
- Arrange the dashboard for operational use.
- Save and share it at the appropriate scope.

AZ-104 exam point: Dashboards provide a consolidated operational view but do not replace logs or alerts.

Scenario 192: Choose RBAC versus Entra role assignment

Scenario / Requirement: Choose RBAC versus Entra role assignment

In simple words: A user needs permission to manage Azure resources, not merely manage identities in Microsoft Entra ID.

Step-by-step solution:

- Identify whether the permission concerns Azure resources or Entra directory objects.

- For Azure resource permissions, evaluate Azure RBAC.
- For directory administration, evaluate the appropriate Entra role.
- Assign only the required role at the appropriate scope.

AZ-104 exam point: Do not confuse Azure RBAC with Microsoft Entra directory roles.

Scenario 193: Choose Policy versus RBAC for mandatory tagging

Scenario / Requirement: Choose Policy versus RBAC for mandatory tagging

In simple words: Every resource must have an Environment tag.

Step-by-step solution:

- Identify that the requirement is a configuration/governance rule.
- Use Azure Policy rather than RBAC.
- Assign the policy at the required scope.
- Test compliance on new and existing resources.

AZ-104 exam point: RBAC controls who can do something; Policy controls whether resource configurations comply with organizational rules.

Scenario 194: Choose a private endpoint for private PaaS access

Scenario / Requirement: Choose a private endpoint for private PaaS access

In simple words: A database must be reachable through a private IP from an application VNet.

Step-by-step solution:

- Identify the requirement for a private IP and private connectivity.
- Choose a private endpoint design.
- Create the endpoint in the application VNet.
- Configure DNS resolution.
- Test from the application subnet.

AZ-104 exam point: The private-IP requirement points toward Private Link/private endpoint rather than a public endpoint restriction.

Scenario 195: Choose NAT Gateway for controlled outbound connectivity

Scenario / Requirement: Choose NAT Gateway for controlled outbound connectivity

In simple words: VMs need outbound internet access without individual public IP addresses.

Step-by-step solution:

- Identify that the requirement is outbound connectivity.
- Associate a NAT Gateway with the required subnet.
- Ensure the workload does not require inbound public access.
- Test outbound connectivity.
- Verify the observed public egress IP.

AZ-104 exam point: NAT Gateway is an outbound connectivity solution, not an inbound load-balancing service.

Scenario 196: Choose Load Balancer for TCP distribution

Scenario / Requirement: Choose Load Balancer for TCP distribution

In simple words: A service needs Layer 4 distribution across multiple VMs.

Step-by-step solution:

- Identify the protocol and load-balancing layer.

- Choose Azure Load Balancer.
- Create frontend, backend pool, probe, and rule.
- Verify backend health and connectivity.

AZ-104 exam point: Match the service to the required network layer.

Scenario 197: Choose Application Gateway for HTTP path-based routing

Scenario / Requirement: Choose Application Gateway for HTTP path-based routing

In simple words: A regional web application needs HTTP/S routing based on URL paths.

Step-by-step solution:

- Identify the need for Layer 7 HTTP/S routing.
- Choose Application Gateway.
- Configure frontend, backend pool, listener, and routing rule.
- Configure health probes.
- Test different URL paths.

AZ-104 exam point: Application Gateway is suited to regional Layer 7 web traffic scenarios.

Scenario 198: Choose Azure Backup versus Site Recovery

Scenario / Requirement: Choose Azure Backup versus Site Recovery

In simple words: One requirement is point-in-time recovery of deleted data; another is rapid regional recovery.

Step-by-step solution:

- For backup/recovery-point requirements, evaluate Azure Backup.
- For regional disaster recovery with replicated workloads, evaluate Site Recovery.
- Configure each service according to its purpose.
- Test the intended recovery operation.

AZ-104 exam point: Backup and disaster recovery solve related but different problems.

Scenario 199: Choose metric alert versus log query

Scenario / Requirement: Choose metric alert versus log query

In simple words: You need an alert when CPU exceeds 90%, but another requirement needs an alert when a specific error event appears in logs.

Step-by-step solution:

- Use a metric alert for the numeric CPU threshold.
- Use a log-based alert for the event/query condition.
- Configure the appropriate evaluation and notification behavior.
- Test both alert paths.

AZ-104 exam point: Choose the monitoring signal that matches the condition.

Scenario 200: Diagnose 'VM is running but application is unreachable'

Scenario / Requirement: Diagnose 'VM is running but application is unreachable'

In simple words: Azure reports the VM as running, but users cannot reach the application.

Step-by-step solution:

- Check whether the application process is running/listening.
- Check the guest OS firewall.

- Check NIC/subnet NSGs.
- Check effective routes.
- Check the load balancer or public IP if one is involved.
- Use Network Watcher to isolate the failing layer.

AZ-104 exam point: A VM power state of Running does not prove application-level connectivity.

Scenario 201: Diagnose 'Storage access denied from VM'

Scenario / Requirement: Diagnose 'Storage access denied from VM'

In simple words: A VM cannot access a storage account even though credentials appear correct.

Step-by-step solution:

- Check storage firewall/network rules.
- Determine whether access uses service endpoint or private endpoint.
- Check DNS for private endpoint scenarios.
- Check identity/RBAC or SAS permissions.
- Test connectivity and authorization separately.

AZ-104 exam point: Separate network reachability problems from authentication/authorization problems.

Scenario 202: Diagnose 'Policy denies deployment'

Scenario / Requirement: Diagnose 'Policy denies deployment'

In simple words: A deployment fails because an organizational rule prevents it.

Step-by-step solution:

- Open the deployment error.
- Identify the policy assignment and denied condition.
- Inspect policy parameters and scope.
- Change the resource configuration if it should comply.
- If the requirement genuinely differs, use the organization's approved policy exception process.
- Retry and verify compliance.

AZ-104 exam point: Policy denial troubleshooting requires identifying the exact policy and condition.

Scenario 203: Diagnose 'RBAC assignment exists but access is denied'

Scenario / Requirement: Diagnose 'RBAC assignment exists but access is denied'

In simple words: A user has a role assignment but still cannot perform the action.

Step-by-step solution:

- Check the assignment scope.
- Check whether the role is inherited to the target resource.
- Check for deny assignments or policy restrictions where relevant.
- Check whether the action requires a different permission.
- Allow for propagation delay and retest.

AZ-104 exam point: An existing role assignment does not automatically mean the requested operation is permitted.

Scenario 204: Diagnose 'Private endpoint exists but hostname resolves publicly'

Scenario / Requirement: Diagnose 'Private endpoint exists but hostname resolves publicly'

In simple words: A client can reach a PaaS service but DNS still resolves to a public address.

Step-by-step solution:

- Check the private endpoint connection.
- Check the private DNS zone.
- Verify the zone is linked to the client VNet.
- Verify the appropriate DNS record exists.
- Test DNS from the client VNet.
- Retest the application connection.

AZ-104 exam point: Private endpoint connectivity often fails because DNS has not been configured correctly.

Scenario 205: Diagnose 'Load Balancer probe is unhealthy'

Scenario / Requirement: Diagnose 'Load Balancer probe is unhealthy'

In simple words: A backend VM is available but the load balancer considers it unhealthy.

Step-by-step solution:

- Check the probe port/protocol.
- Confirm the service is listening on that port.
- Check NSGs and guest firewall.
- Check the health-probe path for HTTP probes.
- Test the endpoint from within the network.
- Correct the issue and recheck health.

AZ-104 exam point: Health probe success is a prerequisite for backend traffic.

Scenario 206: Diagnose 'KQL query returns no results'

Scenario / Requirement: Diagnose 'KQL query returns no results'

In simple words: You enabled diagnostics but cannot find expected logs.

Step-by-step solution:

- Check the diagnostic setting destination.
- Verify the correct log category is enabled.
- Check the correct Log Analytics workspace.
- Expand the time range.
- Confirm the table name and query syntax.
- Generate a new event and query again.

AZ-104 exam point: No query results can be caused by ingestion/configuration, not just KQL syntax.

Scenario 207: Diagnose 'Backup job failed'

Scenario / Requirement: Diagnose 'Backup job failed'

In simple words: A VM backup operation failed and must be restored to a healthy protection state.

Step-by-step solution:

- Open the backup job details.
- Read the failure code/message.
- Check VM agent/connectivity/prerequisites as indicated.
- Check vault and policy configuration.
- Correct the root cause.
- Run an on-demand backup and verify success.

AZ-104 exam point: Use the backup job error details rather than recreating the entire backup configuration blindly.

Scenario 208: Diagnose 'Site Recovery replication unhealthy'

Scenario / Requirement: Diagnose 'Site Recovery replication unhealthy'

In simple words: A protected VM is no longer replicating correctly.

Step-by-step solution:

- Open the replicated item.
- Review replication health and error details.
- Check source VM/network/extension health.
- Check target resource requirements.
- Correct the reported issue.
- Confirm replication returns to a healthy state.

AZ-104 exam point: Replication health must be monitored continuously for effective disaster recovery.

Scenario 209: Exam decision: smallest scope principle

Scenario / Requirement: Exam decision: smallest scope principle

In simple words: A contractor needs read-only access to one resource group for one week.

Step-by-step solution:

- Choose Reader rather than Contributor/Owner.
- Assign it at the resource-group scope rather than subscription scope.
- Use the organization's approved process for temporary access if available.
- Verify the contractor can view but not modify resources.

AZ-104 exam point: Use least privilege and the narrowest practical scope.

Scenario 210: Exam decision: minimize public exposure

Scenario / Requirement: Exam decision: minimize public exposure

In simple words: Administrators need RDP/SSH access to VMs, but security policy prohibits public management ports.

Step-by-step solution:

- Do not expose RDP/SSH directly through public IPs.
- Evaluate Azure Bastion.
- Deploy Bastion in the VNet with the required subnet configuration.
- Connect through the Azure portal and verify management ports are not publicly exposed.

AZ-104 exam point: This tests choosing a service based on a security requirement rather than memorizing a product.

Scenario 211: Exam decision: temporary external access to one blob

Scenario / Requirement: Exam decision: temporary external access to one blob

In simple words: An external partner needs read-only access to one blob for two hours.

Step-by-step solution:

- Avoid giving the partner the storage account key.
- Generate a narrowly scoped read-only SAS for the blob.
- Set a short expiry.
- Share the SAS securely.
- Revoke/rotate the access method if compromise occurs.

AZ-104 exam point: The correct solution should be temporary, scoped, and least-privileged.

Final AZ-104 Coverage Checklist — April 2026 Blueprint

- Microsoft Entra users and groups: creation, properties, licensing, external users, SSPR
- Azure RBAC: built-in roles, scope, inheritance, access interpretation
- Subscriptions, resource groups, management groups, Policy, locks, tags, budgets, Advisor

Storage access: firewall/network rules, SAS, stored access policies, access keys, Azure Files identity-based access

Storage accounts: creation/configuration, redundancy, object replication, encryption, Storage Explorer, AzCopy

- Blob and Files: containers, shares, tiers, soft delete, snapshots, lifecycle, versioning
- ARM/Bicep: interpret, modify, deploy, export, convert, and safely preview infrastructure changes
- VMs: creation, encryption at host, moves, sizing, disks, zones, availability sets, VMSS
- Containers: ACR, ACI, Container Apps, sizing and scaling
- App Service: plan, scaling, app creation, TLS/certificates, custom DNS, backup, networking, deployment slots
- VNets/subnets, peering, public IPs, UDRs, network troubleshooting
- NSGs/ASGs, effective security rules, Bastion, service endpoints, private endpoints
- Azure DNS, internal/public load balancers, load-balancing troubleshooting
- Azure Monitor: metrics, diagnostic/log settings, KQL, alerts, action groups, alert processing rules
- Azure Monitor Insights for VMs, storage, and networks; Network Watcher and Connection Monitor
- Backup: Recovery Services vault, Azure Backup vault, policies, backup/restore, reports and alerts
- Site Recovery: replication, test failover, failover

Cross-cutting Azure administration: portal, Azure CLI, PowerShell, Cloud Shell

- Scenario-based troubleshooting and service-selection practice

Recommended Study Method

- Do each practical scenario in the Azure portal first.
- For CLI/PowerShell scenarios, run the commands in Cloud Shell and verify the result in the portal.
- For troubleshooting scenarios, deliberately create a small fault in a disposable lab and diagnose it before fixing it.
- After completing a scenario, explain in your own words why that Azure service was chosen instead of the closest alternative.
- Use the official Microsoft Learn AZ-104 Skills Measured page as the authoritative checklist for exam scope.